

## ETSA Utilities' Annual Pricing Proposal 2011/12



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**June 2011**



## Shortened forms

| <b>Abbreviation</b>         | <b>Definition or description</b>                                                                                                                                                                                      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AER                         | Australian Energy Regulator.                                                                                                                                                                                          |
| Augmentation                | Investment in new network assets to meet increased demand.                                                                                                                                                            |
| Capacity, capability        | The amount of energy that a part of the network is able to carry.                                                                                                                                                     |
| Capital Contributed Works   | Works for which the customer(s) contribute towards the cost of supplying assets, typically because they are the sole users.                                                                                           |
| COAG                        | Council of Australian Governments.                                                                                                                                                                                    |
| Contestability              | Customer choice of electricity supplier.                                                                                                                                                                              |
| Controlled Load             | The DNSP controls the hours in which the supply is made available.                                                                                                                                                    |
| Cost of Supply Model        | Theoretical and algorithmic model used to calculate prices, which conform to the pricing goals.                                                                                                                       |
| Cross subsidy               | Where the price to a tariff class falls outside the range between the avoidable incremental cost of supply and the cost of stand-alone supply, an economic cross subsidy from or to other customers is said to exist. |
| Decision                    | The Australian Energy Regulator's Final Decision on South Australia - distribution determination 2010–11 to 2014–15, May 2010                                                                                         |
| Demand                      | Energy consumption at a point in time.                                                                                                                                                                                |
| Demand Management           | Attempt to modify demand behaviour so as to constrain demand at critical times.                                                                                                                                       |
| Distribution Network        | The assets and service which links energy customers to the transmission network.                                                                                                                                      |
| Distributor, DNSP           | Distribution Network Service Provider.                                                                                                                                                                                |
| DUoS                        | Distribution Use of System. The utilisation of the distribution network in the provision of electricity to consumers (a component of NUoS).                                                                           |
| EDPD                        | Electricity Distribution Price Determination (South Australian).                                                                                                                                                      |
| ESCoSA                      | Essential Services Commission of South Australia, the South Australian Regulator of energy and other infrastructure.                                                                                                  |
| FRC                         | Full Retail Competition (sometimes called Full Retail Contestability).                                                                                                                                                |
| High Voltage                | Equipment or supplies at voltages of 22 or 11 kV.                                                                                                                                                                     |
| IBT, Inclining Block Tariff | A network tariff energy rate in which the rate increases above specific consumption thresholds.                                                                                                                       |
| kVA, MVA                    | Kilo-volt amps and Mega-volt amps, units of instantaneous total electrical power demand. Usually the peak demand is referenced. See also PF for the relationship between power demand quantities.                     |
| kVAr, MVAR                  | Kilo-volt amps (reactive) and Mega-volt amps (reactive) units of                                                                                                                                                      |

| Abbreviation       | Definition or description                                                                                                                                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | instantaneous reactive electrical power demand. Usually the peak demand is referenced. See also PF for the relationship between power demand quantities.                                                                                                                                                                           |
| kW, MW             | Kilo-watts and Mega-watts, units of instantaneous real electrical power demand. Usually the peak demand is referenced. See also PF for the relationship between power demand quantities.                                                                                                                                           |
| kWh, MWh           | Kilo-watt hours and Mega-watt hours, units of electrical energy consumption.                                                                                                                                                                                                                                                       |
| Low Voltage        | Equipment or supply at a voltage of 220 V single phase or 380 V, three phase.                                                                                                                                                                                                                                                      |
| Marginal Cost      | The cost of providing a small increment of service. The Long Run Marginal Cost (LRMC) includes future investment, Short Run Marginal Cost (SRMC) considers only the costs involved without extra investment.                                                                                                                       |
| Market Participant | Businesses involved in the electricity industry are referred to as Market or Code Participants.                                                                                                                                                                                                                                    |
| Supply Rate        | The fixed daily cost component of a Network price.                                                                                                                                                                                                                                                                                 |
| NEL                | National Electricity Law.                                                                                                                                                                                                                                                                                                          |
| NEM                | National Electricity Market.                                                                                                                                                                                                                                                                                                       |
| NUoS               | Network Use of System. The utilisation of the total electricity network in the provision of electricity to consumers (NUoS = DUoS + TUoS).                                                                                                                                                                                         |
| PF                 | Power Factor, a measure of the ratio of real power to total power of a load. The relationship between real, reactive and total power is as follows:<br>$PF = \frac{\text{Real Power (in kW or MW)}}{\text{Reactive Power (in kVA or MVA)}}$ $\text{Total Power kVA} = \sqrt{\text{Real Power kW}^2 + \text{Reactive Power kVA}^2}$ |
| Price Signal       | Prices set to convey a desired behaviour because of the costs associated with supplying the service.                                                                                                                                                                                                                               |
| Price Structure    | The components that make up a Price available to customers.                                                                                                                                                                                                                                                                        |
| Proposal           | ETSA Utilities' Initial Pricing Proposal, submitted in accordance with the Rules (this document).                                                                                                                                                                                                                                  |
| Retailer           | An FRC market participant (business) supplying electricity to customers.                                                                                                                                                                                                                                                           |
| Rules              | National Electricity Rules.                                                                                                                                                                                                                                                                                                        |
| Subtransmission    | Equipment or supplies at voltage levels of 132, 66 or 33 kV.                                                                                                                                                                                                                                                                       |
| Tariff             | A grouping of customers who are subject to the same network price components and conditions of supply.                                                                                                                                                                                                                             |
| Tariff class       | A class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs.                                                                                                                                                                                                         |
| Transmission       | The assets and service that enable generators to transmit their electrical                                                                                                                                                                                                                                                         |

| <b>Abbreviation</b> | <b>Definition or description</b>                                                                                                                                                                           |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network             | energy to population centres. Operating voltage of equipment is 275, 132 or 66 kV.                                                                                                                         |
| TUoS                | Transmission Use of System charges for the utilisation of the transmission network.                                                                                                                        |
| Unmetered supply    | A connection to the distribution system which is not equipped with a meter and has estimated consumption. Connections to public lights, phone boxes, traffic lights and the like are not normally metered. |
| WAPC                | Weighted Average Price Cap, a form of regulatory price control, where the allowable price change is based on the weighted historic consumption of each price.                                              |
| ToU                 | Time of Use, a system of pricing where energy or demand charges are higher in periods of peak utilisation of the network.                                                                                  |



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## **Pricing Proposal**

### **1 Introduction**

ETSA Utilities is submitting this annual Pricing Proposal (Pricing Proposal) for 2011/12 to the Australian Energy Regulator (AER). The proposal has been prepared in accordance with the requirements of the National Electricity Rules (Rules), the AER's 2010 regulatory determination and subsequent amendments to the determination made by the Australian Competition Tribunal.

The Pricing Proposal covers all of ETSA Utilities' direct control services. In its 2010 South Australian Decision (Decision) the AER further classified these direct control services as standard control services and alternative control services<sup>1</sup>.

#### **1.1 National Electricity Rules**

Clause 6.1.1 of the Rules confers responsibility on the AER for the economic regulation of distribution services provided by means of, or in connection with, distribution systems that form part of the national grid.

##### **1.1.1 Classification of distribution services**

In accordance with clause 6.2.1 and 6.2.2 of the Rules, the AER has classified ETSA Utilities' distribution services into the following three classes:

1. Direct control services;
2. Negotiated distribution services;
3. Unregulated distribution services.

Direct control services have been further divided into the following two subclasses:

1. Standard control services; and
2. Alternative control services.

This Pricing Proposal applies to all of the direct control services provided by ETSA Utilities.

#### **1.2 Scope of ETSA Utilities' Pricing Proposal**

ETSA Utilities' Pricing Proposal sets out the proposed prices required to comply with the Weighted Average Price Cap approved by the Australian Energy Regulator (AER) in the 2010 determination.

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<sup>1</sup> Australian Energy Regulator, Final Decision - South Australia distribution determination 2010–11 to 2014–15, May 2010.

This Pricing Proposal is submitted in accordance with, and complies with, the requirements of:

1. The National Electricity Law;
2. The National Electricity Rules, including the Transitional Rules for South Australia;
3. The AER's 2010-15 South Australian distribution determination, including the AER's Final Decisions on the control mechanisms and pricing X factors for standard control services and alternative control services; and
4. The Australian Competition Tribunal's decision on 19 May 2011, to vary the AER determination by permitting the inclusion of easements within the Regulatory Asset Base (RAB) and varying the corporate taxation allowance.

### **1.3 Structure of ETSA Utilities' Pricing Proposal**

In Part I of the Rules, clause 6.18 sets out the requirements concerning Distribution Pricing. These requirements include:

- The Pricing Principles which must be followed;
- The requirement for this Pricing Proposal; and
- The matters that the Pricing Proposal must address.

The AER's determination has been made pursuant to clause 6.11.1 of the Rules. Several aspects of that determination impose requirements concerning distribution pricing, including:

- Classification of services;
- The pricing control mechanism(s), X factors and side constraints;
- Assigning and reassigning customers to tariff classes;
- Recovery of transmission charges; and
- The passthrough of costs, in specified circumstances.

This Pricing Proposal has been structured so as to allow compliance with the specific requirements of the Rules and the AER's determination to be readily ascertained. The substantive sections of this Pricing Proposal are set out in Table 1.

**Table 1 - Structure of ETSA Utilities' Pricing Proposal**

| Chapter                                 | Purpose                                                                                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 Regulatory Requirements               | Summarises the regulatory requirements as they relate to ETSA Utilities' Pricing Proposal including the relevant requirements of the Rules and the AER's 2010 determination                                               |
| 3 Business Overview                     | Summarises the characteristics of ETSA Utilities' network that provide the context for ETSA Utilities' network tariff strategy.                                                                                           |
| 4 Tariff Classes                        | Defines the tariffs and tariff classes into which ETSA Utilities' customers for direct control services are divided and their charging parameters.                                                                        |
| 5 Network Tariff Strategy               | Outlines ETSA Utilities' Network Tariff Strategy and indicates how tariff charging parameters are expected to vary.                                                                                                       |
| 6 Standard Control Services Tariffs     | Describes the nature and extent of the change in ETSA Utilities' DUoS tariffs between 2010/11 and 2011/12.                                                                                                                |
| 7 Customer Impacts                      | Outlines the expected customer impacts of ETSA Utilities' NUoS prices in 2011/12 and the system of reviewing those impacts throughout the regulatory control period.                                                      |
| 8 Pricing of standard control services  | Demonstrates that ETSA Utilities' 2011/12 prices comply with the pricing X factors, side constraints and the NER pricing principles.                                                                                      |
| 9 Transmission Cost Recovery Tariffs    | Sets out ETSA Utilities' Transmission cost recovery tariff setting methodology and demonstrates adjustments made to the tariffs resulting from the actual and estimated recovery of these charges in 2009/10 and 2010/11. |
| 10 Customer Assignment and Reassignment | Sets out ETSA Utilities' tariff assignment and reassignment strategy.                                                                                                                                                     |
| 11 Alternative control services         | Sets out ETSA Utilities' tariffs for alternative control metering services.                                                                                                                                               |
| Appendices                              | Separately provided.                                                                                                                                                                                                      |

## 1.4 Confidential information

Clause 6.19.2 of the Rules classifies as confidential all network pricing information about a Distribution Network User used by a DNSP for the purposes of network pricing. ETSA Utilities has nominated some of the attachments that constitute parts of this Pricing Proposal as confidential.

ETSA Utilities requests that the AER does not disclose the information contained in these confidential attachments to any person outside of the AER.

## 2 Regulatory requirements

This Chapter summarises the regulatory requirements pertaining to ETSA Utilities' Pricing Proposal, including the relevant requirements of the Rules and those of the AER's 2010 distribution determination.

### 2.1 Rules requirements

#### 2.1.1 Required elements of a Pricing Proposal

To comply with clause 6.18.2 of the Rules, ETSA Utilities' 2012 Pricing Proposal must include the elements below<sup>2</sup>:

#### 6.18.2 Pricing proposals

(a) A *Distribution Network Service Provider* must:

...

- (2) submit to the *AER*, at least 2 months before the commencement of the second and each subsequent *regulatory year* of the *regulatory control period*, a further *pricing proposal* (an "annual pricing proposal") for the relevant *regulatory year*.

(b) A *pricing proposal* must:

- (1) set out the *tariff classes* that are to apply for the relevant *regulatory year*; and
- (2) set out the proposed tariffs for each *tariff class*; and
- (3) set out, for each proposed tariff, the *charging parameters* and the elements of service to which each *charging parameter* relates; and
- (4) set out, for each *tariff class* related to *standard control services*, the expected weighted average revenue for the relevant *regulatory year* and also for the current *regulatory year*; and
- (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the *regulatory year* and the basis on which it could occur; and
- (6) set out how charges incurred by the *Distribution Network Service Provider* for *transmission use of system services* are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous *regulatory year*; and
- (7) demonstrate compliance with the *Rules* and any applicable distribution determination; and
- (8) describe the nature and extent of change from the previous *regulatory year* and demonstrate that the changes comply with the *Rules* and any applicable distribution determination.

<sup>2</sup> Australian Energy Market Commission, National Electricity Rules, version 35, 25 March 2010.

In accordance with clause 6.18.2(a) of the Rules, ETSA Utilities is submitting this Pricing Proposal for the second year of the 2010-15 regulatory control period to the AER within the required period after publication of the distribution determination.

This Pricing Proposal has been prepared by ETSA Utilities in such a way as to demonstrate in a logical sequence that it complies with all of the requirements of clause 6.18.2(b) of the Rules above.

The other relevant sections of the Rules that have been addressed in formulating this Pricing Proposal are as follows:

- 6.18.3 Tariff classes
- 6.18.5 Pricing principles
- 6.18.6 Side constraints on tariffs for standard control services
- 6.18.7 Recovery of charges for transmission use of system services
- 6.18.8 Approval of pricing proposal
- 6.18.9 Publication of information about tariffs and tariff classes

Reference to these clauses has been made in the appropriate sections of this Pricing Proposal, to demonstrate how ETSA Utilities has complied with each applicable Rules provision.

## **2.2 Requirements of the AER's 2010 Determination**

The AER's 2010 Decision for South Australia has been made pursuant to the provisions clause 6.11.1 of the Rules and imposes a number of requirements that are relevant to a Pricing Proposal. The relevant requirements are as follows:

- Chapter 2 Classification of services (including Appendix A)
  - Assigning customers to tariff classes (Appendix B)
- Chapter 4 Control mechanism for standard control services
  - Changes to tariff structures (Appendix E)
  - Transmission use of system under and overs account (Appendix F)
- Chapter 12 STPIS arrangements (Appendix K)<sup>3</sup>
- Chapter 13 EBSS arrangements (Appendix K)<sup>3</sup>
- Chapter 14 DMIS arrangements (Appendix K)<sup>3</sup>
- Chapter 15 Pass through arrangements
- Chapter 16 Pricing X factors
- Chapter 17 Alternative control services

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<sup>3</sup> This requirement involves an adjustment to prices that is not applicable in 2011/12, the second year of the 2010-15 regulatory control period.

Where it is necessary to demonstrate that ETSA Utilities has complied with a requirement of the AER's Decision, reference to the relevant component of the Decision has been made in the appropriate section of this Pricing Proposal.

## 2.3 Principal elements of the AER's 2010 Decision

The principal elements of the AER's Decision pertaining to direct control services (comprising standard and alternative control services) are outlined in this section.

### 2.3.1 Weighted Average Price Cap for standard control services

In Chapter 4 of the 2010 South Australia Decision, the AER has determined the WAPC formula to apply to ETSA Utilities' standard control services for the next regulatory control period<sup>4</sup>. This is as follows:

$$(1+CPI_t)' (1-X_t)' (1+S_t)' (1+D_t)' (1+U_t)' (1+EDPD_t) \pm (\text{passthrough}_t)^3 \frac{\prod_{i=1}^n \prod_{j=1}^m p_t^{ij} q_{t-2}^{ij}}{\prod_{i=1}^n \prod_{j=1}^m p_{t-1}^{ij} q_{t-2}^{ij}}$$

Where:

ETSA Utilities has 'n' distribution tariffs, which each have up to 'm' distribution tariff components, and where:

regulatory year  $t$  is the regulatory year in respect of which the calculation is being made

regulatory year  $t-1$  is the regulatory year immediately preceding regulatory year  $t$

regulatory year  $t-2$  is the regulatory year immediately preceding regulatory year  $t-1$

$p_t^{ij}$  is the proposed distribution tariff for component  $j$  of distribution tariff  $i$  in regulatory year  $t$

$p_{t-1}^{ij}$  is the distribution tariff being charged in regulatory year  $t-1$  for component  $j$  of distribution tariff  $i$

$q_{t-2}^{ij}$  is the quantity of component  $j$  of distribution tariff  $i$  that was delivered in regulatory year  $t-2$

$X_t$  is the allowed real change in average prices from year  $t-1$  to year  $t$  of the regulatory control period as determined by the AER

$S_t$  is the Service Target Performance Incentive Scheme factor to be applied in regulatory year  $t$

$D_t$  is the demand management incentive scheme factor to be applied in regulatory year  $t$ , which is set equal to zero for each year of the next regulatory control period

$U_t$  is the undergrounding factor to be applied in regulatory year  $t$

<sup>4</sup> AER, Final Decision - South Australia distribution determination 2010–11 to 2014–15, May 2010, pp. 25–26.

$EDPD_t$  is the EDPD transition factor for regulatory year  $t$ . It is a carryover of adjustments made in the 2005–2010 EDPD comprising the previous K, Q, PU and SI factor adjustments

$passthrough_t$  is the change in approved pass through amounts, expressed in percentage form, with respect to regulatory year  $t$  as compared to regulatory year  $t-1$ , as determined by the AER

$CPI_t$  is the annual percentage change in the ABS Consumer Price Index All Groups, Weighted Average of Eight Capital Cities from March in regulatory year  $t-2$  to March in regulatory year  $t-1$ .

## 2.3.2 Side constraint for standard control services

Chapter 4 of the AER Decision also contains the side constraint formula to apply to ETSA Utilities' standard control services for the next regulatory control period<sup>5</sup>:

$$(1+CPI_t)' (1-X_t)' (1+S_t)' (1+D_t)' (1+U_t)' (1+EDPD_t)' (1+2\%) \pm (passthrough_t)^3 \frac{\prod_{j=1}^m d_t^j \cdot q_{t-2}^j}{\prod_{j=1}^m d_{t-1}^j \cdot q_{t-2}^j}$$

Where each tariff class ' $j$ ' has up to ' $m$ ' components and where, in addition to the symbols defined in section 2.3.1:

$d_t^j$  is the proposed price for component  $j$  of the tariff class for year  $t$

$d_{t-1}^j$  is the price charged by the DNSP for component  $j$  of the tariff class in year  $t-1$

$q_{t-2}^j$  is the audited quantity of component  $j$  of the tariff class that was charged by the DNSP in year  $t-2$

$X_t$  is the allowed real change in average prices from year  $t-1$  to year  $t$  of the regulatory control period as determined by the AER. If  $X > 0$ , then  $X$  will be set equal to zero for the purposes of the side constraint formula

In addition, ETSA Utilities can not raise the fixed supply charge for small customers by more than \$10 per annum during the next regulatory control period.

## 2.3.3 Revenue requirement and pricing X factors for standard control services

Chapter 16 of the Decision contained ETSA Utilities' revenue requirements and pricing X factors for standard control services. This determination was subsequently amended by orders made by the Australian Competition Tribunal (Tribunal), in respect of:

- The inclusion of easement values in ETSA Utilities' opening asset value for the regulatory control period; and
- The allowance for corporate taxation during the regulatory control period<sup>6</sup>.

<sup>5</sup> AER, Final Decision, May 2010, p. 27.

<sup>6</sup> Australian Competition Tribunal, Application under Section 71b of the National Electricity Law for a Review of a Distribution Determination made by the Australian Energy Regulator in relation to ETSA Utilities, 19 May 2011, p. 4.

Table 2 summarises the revised annual revenue requirements and pricing X factors for the 2010-15 regulatory control period, for standard control services. The main effect of the Tribunal's decision was to increase the X factor in the 2011/12 year.

**Table 2 - Annual revenue requirements and X factors for standard control services (\$m, nominal)**

|                             | 2010/11 | 2011/12 | 2012/13 | 2013/14 | 2014/15 |
|-----------------------------|---------|---------|---------|---------|---------|
| Regulatory depreciation     | 97.0    | 110.0   | 123.4   | 139.1   | 154.2   |
| Return on capital           | 283.0   | 304.8   | 330.9   | 352.6   | 373.8   |
| Operating expenditure       | 198.0   | 209.7   | 221.9   | 237.5   | 248.8   |
| Tax allowance               | 80.7    | 81.8    | 80.3    | 84.1    | 86.7    |
| Capex carryover             | 8.6     | 7.9     | 4.5     | 0.4     | 0.0     |
| Annual revenue requirements | 667.3   | 714.2   | 761.1   | 813.7   | 863.4   |
| Expected revenues           | 618.8   | 726.6   | 776.7   | 841.7   | 868.0   |
| Forecast CPI (%)            | 2.52    | 2.52    | 2.52    | 2.52    | 2.52    |
| X factors (%) <sup>d</sup>  | -12.14  | -18.10  | -7.00   | -7.00   | -0.89   |

(d) Negative values for X indicate real price increases under the CPI-X formula.

The associated pricing X factors for standard control services have been incorporated into this Pricing Proposal.

### 2.3.4 Weighted Average Price Cap for alternative control services

In Chapter 17 of the Decision, the AER has determined the following weighted average price cap formula for ETSA Utilities' alternative control services<sup>7</sup>:

$$(1 + CPI_t)' (1 - X_t)^3 \frac{\sum_{i=1}^n \sum_{j=1}^m p_t^{ij} q_{t-2}^{ij}}{\sum_{i=1}^n \sum_{j=1}^m p_{t-1}^{ij} q_{t-2}^{ij}}$$

Where ETSA Utilities has  $n$  distribution tariffs, which each have up to  $m$  distribution tariff components and where, in addition to the symbols defined in section 2.3.1:

$X_t$  is to be determined using the building block approach.

### 2.3.5 Revenue requirement and pricing X factors for alternative control services

Chapter 17 of the Decision also outlines the AER's final determination on ETSA Utilities' annual revenue requirements and pricing X factors for the 2010-15 regulatory control period for standard control services<sup>8</sup>. The Australian Competition Tribunal's decision on ETSA

<sup>7</sup> AER, Final Decision - South Australia distribution determination 2010-11 to 2014-15, May 2010, pp. 273-274.

<sup>8</sup> AER, Final Decision, May 2010, p. 272.

Utilities' corporate taxation allowance also varied the annual revenue requirement for alternative control services<sup>9</sup>.

The revised decision is summarised in Table 3.

**Table 3 - Annual revenue requirements and X factors for alternative control services (\$m, nominal)**

|                                         | 2010/11 | 2011/12 | 2012/13 | 2013/14 | 2014/15 |
|-----------------------------------------|---------|---------|---------|---------|---------|
| Regulatory depreciation                 | 7.87    | 8.71    | 9.54    | 10.20   | 10.90   |
| Return on capital                       | 3.60    | 4.36    | 5.19    | 6.03    | 6.98    |
| Operating expenditure                   | 6.49    | 6.71    | 7.05    | 7.44    | 7.86    |
| Tax allowance                           | 1.20    | 1.38    | 1.58    | 1.77    | 1.98    |
| Annual revenue requirement <sup>a</sup> | 19.17   | 21.16   | 23.36   | 25.43   | 27.72   |
| X factors <sup>d</sup> (%)              | -8.05   | -16.50  | -6.90   | -6.90   | -3.03   |

(a) Unsmoothed revenue requirement.

(d) Negative values for X indicate real price increases under the CPI-X formula.

## 2.4 Publication of information about tariffs and tariff classes

Clause 6.18.9 of the Rules requires ETSA Utilities to publish the following information on its tariffs and tariff classes.

### 6.18.9 Publication of information about tariffs and tariff classes

- (a) A *Distribution Network Service Provider* must maintain on its website:
- (1) a statement of the provider's *tariff classes* and the tariffs applicable to each class; and
  - (2) for each tariff – the *charging parameters* and the elements of the service to which each *charging parameter* relates;
  - (3) a statement of expected price trends (to be updated for each *regulatory year*) giving an indication of how the *Distribution Network Service Provider* expects prices to change over the *regulatory control period* and the reasons for the expected changes.
- (b) The information for a particular *regulatory year* must, if practicable, be posted on the website 20 *business days* before the commencement of the relevant *regulatory year* and, if that is not practicable, as soon as practicable thereafter.

The information on tariffs and tariff classes contained in the following sections of this Pricing Proposal has been prepared and published in conformity with the requirements of this clause.

<sup>9</sup> Australian Competition Tribunal, 19 May 2011, pp. 8-9.

### **3 Business overview**

This chapter of the Pricing Proposal provides contextual information on ETSA Utilities' business circumstances. This provides the backdrop both to ETSA Utilities' existing network tariffs and the rationale for the changes to tariffs that are proposed during the 2010-15 regulatory control period.

#### **3.1 ETSA Utilities' business**

ETSA Utilities' distribution network covers the State of South Australia, a vast territory of about 178,200 square km, with a coastline of over 5,000 km.

The network's route length extends to more than 85,000 km, with approximately 18% of that length underground. The network includes 393 substations, 69,000 distribution transformers, 723,000 poles and 1.1 million meters.

The extent of ETSA Utilities' operations in South Australia is shown in Figure 1. The South Australian distribution network is predominantly three-phase, with a single-phase system used mostly in rural and remote areas. A sub-transmission network supplies and links zone substations, and operates at 66 kiloVolts (kV) and 33 kV. In rural and remote areas the single-phase system operates at 19 kV. Overall, some 30% of the network is comprised of these long 'single wire earth return' (SWER) lines.

In higher density rural and urban locations, the three-phase feeder system operates at 11 kV. The standard low voltage customer supply is 230V at 50Hz.

#### **3.2 State-wide pricing**

The South Australian government has imposed a requirement on ETSA Utilities to maintain state-wide pricing for small customers (with annual consumption not exceeding 160 MWh)<sup>10</sup>. As a consequence, all of ETSA Utilities' distribution tariffs are averaged. For larger business customers with energy consumption in excess of 40 GWh or a demand greater than 10 MW, locational transmission use of system (TUoS) charges apply.

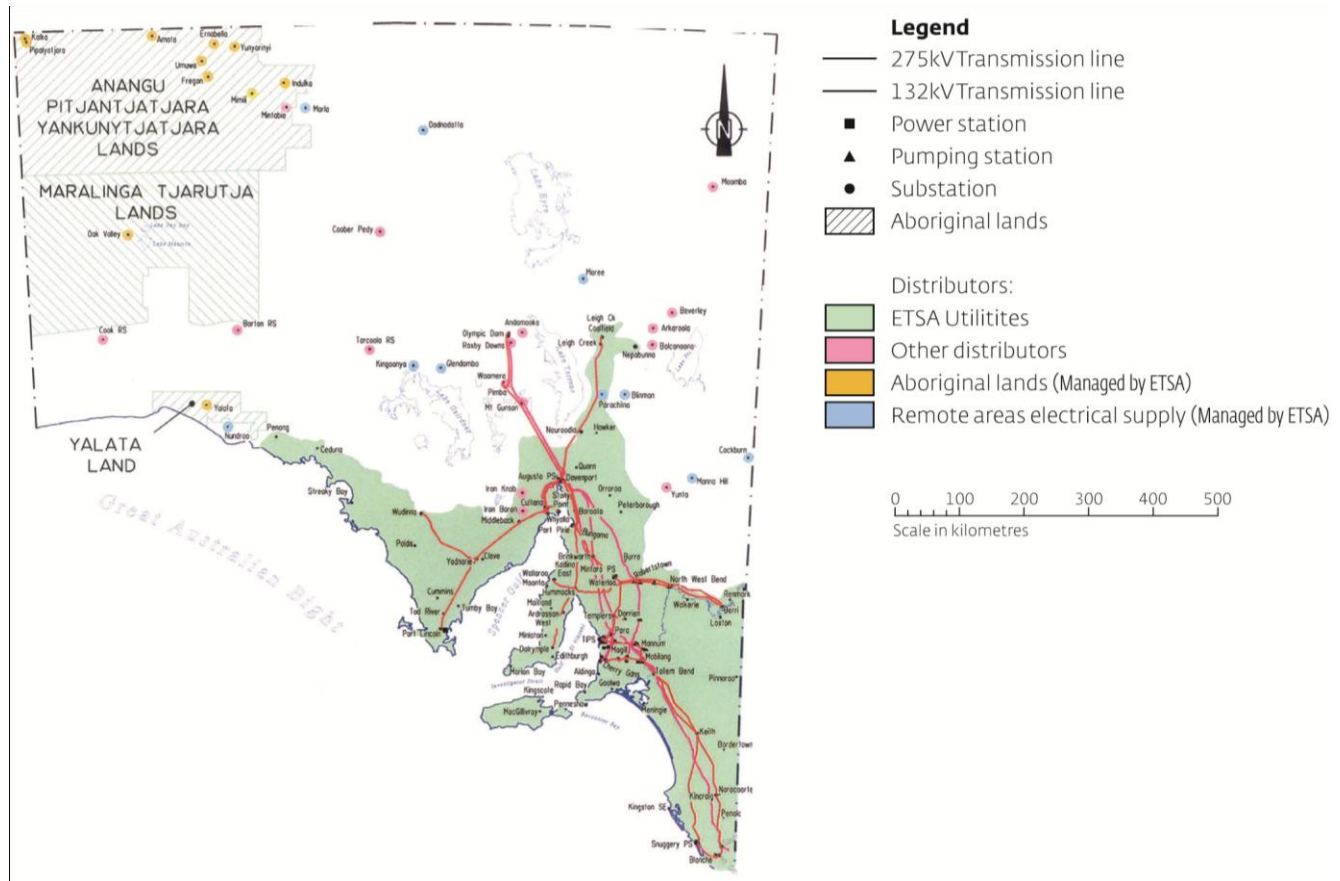
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<sup>10</sup> South Australian Treasurer, Electricity Act 1996 Section 35B Electricity Pricing Order, 11 October 1999.

## 3.3 Characteristics of the region

With the exception of much of the coastal area and the hinterland, South Australia is very sparsely settled. The area serviced by ETSA Utilities' distribution system is shown in Figure 1.

**Figure 1 - ETSA Utilities' service area**



Approximately 70% of customers, including the great majority of business and commercial customers, reside in Adelaide. However, the extensive area serviced by distribution means that 70% of the network infrastructure is required to deliver energy to the remaining 30% of customers. Compared with other states, there are relatively few regional centres, and they are generally small and located widely across the territory.

As a result, the average customer density across the State is very low.

## 3.4 Climactic conditions

Adelaide and much of South Australia has a dry climate featuring greater extremes of summer temperature than most other Australian capitals. Extended periods of heatwave conditions can occur in summer (March 2008 and January 2009 have been the most recent).

During these heatwave periods, summer daytime temperatures can exceed 40°C for several days in a row.

## 3.5 ETSA Utilities' customer and demand profile

The South Australian climate has led to an extraordinary demand for air conditioning. Approximately 90% of homes are now air conditioned, but the consequent high peak network demand occurs for only a small part of the year.

Extremely 'peaky' conditions such as these require network assets and capacity that is under-utilised during much of the year, driving distribution costs higher, on a per unit of energy served basis, than comparable interstate networks.

These conditions also provide the impetus for ETSA Utilities' network tariff strategies and innovative tariff developments described in sections 5.4 to 5.8.

## 4 Tariff Classes

This section describes ETSA Utilities' standard control service tariff classes and the way in which they have been constituted to comply with the requirements of the Rules and the AER's Determination.

In the Appendix A of the Decision, AER has classified the following distribution services provided by ETSA Utilities as standard control services<sup>11</sup>:

### A.1 'Standard' network services

- a. All *network services* except:
  - i. *network services* provided at the request of a *distribution network user*:
    - 1. with higher quality or reliability standards, or lower quality or reliability standards (where permissible), than are required by the NER, the *Electricity Distribution Code*, or any other applicable regulatory instruments, or
    - 2. in excess of levels of service or plant ratings required to be provided by ETSA Utilities' assets, or
  - ii. extension or augmentation of the *distribution network* associated with the provision of a new *connection point* or upgrading of the capability of a connection point to the extent that a *distribution network user* is required to make a financial contribution in accordance with the *Electricity Distribution Code*, or
  - iii. other *network services* that are classified as *negotiated distribution services* in sections B.7 to B.16 of this appendix B.

### A.2 'Standard' connection services

- a. All *connection services* except:
  - i. *connection services* provided at the request of a *distribution network user*:
    - 1. with higher quality or reliability standards, or lower quality or reliability standards (where permissible), than are required by the NER, the *Electricity Distribution Code*, or any other applicable regulatory instrument, or
    - 2. in excess of levels of service or plant ratings required to be provided by ETSA Utilities' assets, or
  - ii. the provision of a new *connection point* or upgrading of the capability of a connection point to the extent a *distribution network user* is required to make a financial contribution in accordance with the *Electricity Distribution Code*, or
  - iii. other *connection services* that are classified as *negotiated distribution services* in sections B.7 to B.16 of this appendix B.

### A.3 Fixed 'standard' 'small' customer metering services

- a. The provision of *energy data services* in respect of meters meeting the requirements of a *metering installation* type 6 except the quarterly meter read service.

<sup>11</sup> AER, Final Decision - South Australia distribution determination 2010–11 to 2014–15, May 2010, pp. 280-281.

## A.4 Unmetered metering services

- a. The provision of metering services in respect of meters meeting the requirements of a *metering installation* type 7.

This section of the Pricing Proposal sets out ETSA Utilities' approach to the pricing of these standard control services and demonstrates compliance with the Rules and the AER's determination.

The equivalent information for the separate tariff class associated with ETSA Utilities' alternative control metering services is included in this Pricing Proposal in Chapter 11.

### 4.1 Regulatory requirements

#### 4.1.1 Rule requirements

ETSA Utilities' Pricing Proposal must contain the information on tariffs, tariff classes and charging parameters set out in clause 6.18.2 of the Rules.

#### 6.18.2 Pricing proposals

- (b) A *pricing proposal* must:
  - (1) set out the *tariff classes* that are to apply for the relevant *regulatory year*; and
  - (2) set out the proposed tariffs for each *tariff class*; and
  - (3) set out, for each proposed tariff, the *charging parameters* and the elements of service to which each *charging parameter* relates;

ETSA Utilities is required to comply with the following requirements of clause 6.18.3 of the Rules with respect to tariff classes.

#### 6.18.3 Tariff classes

- (a) A *pricing proposal* must define the *tariff classes* into which customers for *direct control services* are divided.
- (b) Each customer for *direct control services* must be a member of 1 or more *tariff classes*.
- (c) Separate *tariff classes* must be constituted for customers to whom *standard control services* are supplied and customers to whom *alternative control services* are supplied (but a customer for both *standard control services* and *alternative control services* may be a member of 2 or more *tariff classes*).
- (d) A *tariff class* must be constituted with regard to:
  - (1) the need to group customers together on an economically efficient basis; and
  - (2) the need to avoid unnecessary transaction costs.

## 4.1.2 Requirements of the AER's Decision

The AER accepted ETSA Utilities' proposal that standard control services customer tariffs would be classified into four tariff classes<sup>12</sup>.

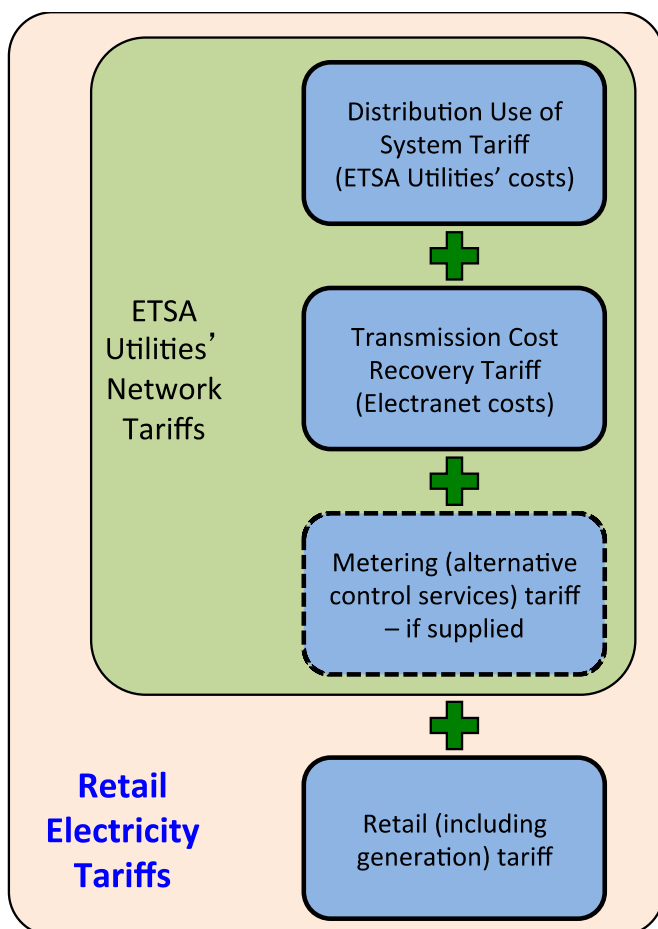
The AER has established procedures for assigning or reassigning customers to tariff classes in Appendix B of its Decision. At the commencement of the regulatory control period, all existing customers were assigned to the tariff class that ETSA Utilities had assigned prior to 1 July 2010<sup>13</sup>.

Appendix B of the AER's Decision contains procedures for the reassigning of customers to tariff classes, with which ETSA Utilities must comply during the 2010-15 regulatory control period. These procedures are set out in section 10.1.2.

## 4.2 Standard control service tariffs and tariff classes

ETSA Utilities' network use of system tariffs are an aggregation of distribution use of system tariffs, metering service tariffs and transmission cost recovery tariffs. The components of these tariffs are illustrated in Figure 2.

**Figure 2 - Components of ETSA Utilities network tariffs**



Retailers may pass through the components of ETSA Utilities' network tariffs to customers directly, or modify their structure by bundling with the retail component, which includes the cost of purchasing generated energy from the NEM, plus retail costs.

This section outlines the distribution tariff arrangements, which are designed to recover the cost of providing prescribed control services to customers. These services are segregated into tariffs and tariff classes, which cover all of direct control services that ETSA Utilities provides, as required by clauses 6.18.3(a) and 6.18.3(b).

Section 11 of this Pricing Proposal outlines the arrangements for ETSA Utilities' alternative control metering service tariffs, which in accordance with clause 6.18.3(c) of the Rules has been constituted as a separate tariff class

<sup>12</sup> AER, Final Decision, May 2010, p. 23.

<sup>13</sup> AER, Final Decision, May 2010, p. 287.

with separate charging parameters.

The transmission cost recovery section 9 describes how the transmission costs incurred by ETSA Utilities are recovered from customers, again with separate charging parameters.

The grouping of customers into standard control service tariffs has historically distinguished between customers on the basis of the following factors:

- The nature and extent of usage of different types of customer;
- For business customers, nature of connection to the network, including the capacity and location or voltage of connection;
- Whether the customer opts to have quarterly or monthly billing;
- Whether the customer also receives a controlled load service; and
- The type of meter installed at the premises, with a distinction between Types 1-4 metering and Types 5-6 metering.

It should be noted that ETSA Utilities uses net metering for all customer billing and does not distinguish between customers with micro-generation and those without, in either the network tariff or network tariff class.

An important consideration in establishing this set of tariff classes in 2010 was to reduce the complexity of the overall arrangement by grouping customer tariffs with a similar connection and usage profile together on an economically efficient basis and thereby avoiding unnecessary transaction costs. Thus, for example, there are six tariffs for the primary supply to residential customers, all of which have identical charging parameters and price levels.

Clearly, in establishing tariff classes that are to be used for the purpose of monitoring pricing compliance, it is desirable and appropriate that similar individual tariffs should be grouped together. This is particularly the case for some business tariffs, where one or a few large customers would dominate the class and the side constraint would not apply to a tariff class but those large customers.

ETSA Utilities' network tariffs and tariff classes for 2011/12 are shown in Figure 3 below. The tariff classes have been constituted with regard to the provisions of clause 6.18.3(d) of the Rules concerning economic efficiency and transaction costs. As is apparent from the diagram, the suite of tariffs provides:

- A range of tariffs which are dependent upon a customer's size, consumption characteristics and voltage of connection (these factors are generally related); and
- A level of sophistication in the pricing and billing, in some cases facilitated by the metering arrangements.

A number of obsolete tariffs (principally business kW demand tariffs) existed at the time of the AER's 2010 determination. Subsequently, all customers were transferred from those tariffs by the commencement of the 2010/11 year.

**Figure 3 - ETSA Utilities' existing network tariffs**

| Type 1-4 meter<br>Monthly billing                                                                                             | Type 5,6 meter           |                          | Type 7<br>(unmetered)<br>Monthly billing |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|------------------------------------------|
| <b>Major business (11, 33, 66 kV)</b><br>kVA demand (locational TUoS)<br>kVA demand (loc'l TUoS) >10MW<br>kVA demand Zone ZVS |                          |                          |                                          |
| <b>High voltage business (11 kV)</b><br>kVA demand VHVS<br>kVA demand VHLVS (<1000kVA)<br>kVA demand VHVO<br>2 rate B2R124HV  |                          |                          |                                          |
| <b>Low voltage business</b>                                                                                                   |                          |                          |                                          |
| kVA demand VLVS                                                                                                               |                          |                          |                                          |
| 2 rate B2R124                                                                                                                 | 2 rate MB2R              | 2 rate QB2R              |                                          |
| BSR124                                                                                                                        | MBSR                     | QBSR                     | LVUU                                     |
| With cont. load BSR124OPCL                                                                                                    | With cont. load MBSROPCL | With cont. load QBSROPCL | LVUU24                                   |
| <b>Low voltage residential</b>                                                                                                |                          |                          |                                          |
| MRSRI                                                                                                                         | MRSR                     | QRSR                     |                                          |
| With cont. load MRSRCLI                                                                                                       | With cont. load MRSROPCL | With cont. load QRSROPCL |                                          |

#### 4.2.1 Standard control services tariffs

The 25 tariffs in Figure 3 cover a diverse range of customer requirements.

Residential customer tariffs have a fixed daily charge (termed the Supply Rate) and an inclining block energy component, in common with the tariff structures of many utilities. In 2009/10, the inclining block energy charge was modified to include four block levels. A separate energy rate applies to the energy consumption within each block level. Each of the tariff components (charging parameters) is determined in accordance with the WAPC price control formula.

Business customer tariffs cover the range of:

- Inclining block energy tariffs for small Low Voltage connected customers;
- Two rate Time of Use (peak and off peak) for larger Low Voltage connected customers; and
- kVA demand tariffs for the largest customers at all voltage levels, with individually calculated (locational) transmission prices for those customers with a demand in excess of 10 MW. The demand charge is stepped according to the customer's size, to reflect the cost of providing network capacity at the particular voltage level and location.

## **4.2.2     *Standard control services tariff classes***

The four tariff classes that ETSA Utilities has established are as follows:

- Major business (11 to 66 kV);
- High Voltage business
- Low voltage business including unmetered supplies; and
- Residential.

ETSA Utilities has illustrated the grouping of its individual tariffs into tariff classes in Figure 3 using red outlines. For clarity, this illustration does not include the current range of obsolete tariffs, most of which are expected to be able to be withdrawn as customers migrate to standard tariffs before the commencement of the 2010-15 regulatory control period.

A description of the tariffs in each of the tariff classes and their charging parameters follows. More detailed information on the application of the tariffs is available from ETSA Utilities' web site. This section does not describe the range of obsolete tariffs.

Note that, for completeness, those components of charging parameters associated with transmission recovery tariffs and alternative control services have been shown in the following section.

## **4.3        *Low voltage residential tariff class***

This tariff class includes the residential single rate and controlled load tariffs.

### **4.3.1     *Low voltage residential single rate tariff***

The low voltage residential single rate tariff is available to eligible residential customers taking supply at less than 1 kV. These customers ordinarily use a type 1-6 NEM compliant meter and metered energy consumption is charged in four blocks.

The low voltage residential single rate tariff incorporates the charging parameters set out in Table 4.

**Table 4 - Low voltage residential tariff single rate charging parameters**

| Charging Parameter | Units  | Element of service  |                            |                              |                                                     |
|--------------------|--------|---------------------|----------------------------|------------------------------|-----------------------------------------------------|
|                    |        | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                         |
| Supply Rate        | \$/day | ✓                   | -                          | If applicable                | Fixed daily charge                                  |
| Block 1 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For consumption up to and including 333.3 kWh/month |
| Block 2 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the next 500 kWh/month                          |
| Block 3 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the next 833.3 kWh/month                        |
| Block 4 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the balance of consumption                      |

A negotiated service charge will apply where ETSA Utilities is required to read the type 1-4 meter (eg. for tier one customers and for tier two customers < 160MWh pa).

This tariff is invoiced monthly or quarterly. A negotiated distribution service charge also applies for the monthly reading of Type 5-6 meters.

#### 4.3.2 Controlled load tariff

The low voltage controlled load tariff is available for permanently installed storage water heaters with a rated delivery of not less than 125 litres, storage space heaters and other approved applications. A time switch for the control of the heater and separate metering is installed.

The controlled load tariff incorporates the charging parameters set out in Table 5.

**Table 5 - Controlled load tariff charging parameters**

| Charging Parameter | Units  | Element of service  |                            |                              |                                                     |
|--------------------|--------|---------------------|----------------------------|------------------------------|-----------------------------------------------------|
|                    |        | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                         |
| Block 1 Usage Rate | \$/kWh | ✓                   | ✓                          |                              | For consumption up to and including 666.7 kWh/month |
| Block 2 Usage Rate | \$/kWh | ✓                   | ✓                          |                              | For the balance of consumption                      |

This tariff is available only to customers that were taking supply under the controlled load tariff as at 30 June 2003, or are utilising a business single or residential tariff at the NMI in

conjunction with the controlled load. This tariff is invoiced at the same frequency as the primary tariff.

## 4.4 Low voltage business tariff class

The low voltage business tariffs cover a broad range of customer sizes and types of metering installations. This tariff class is the most diverse in terms of its consumption range and the variety of end uses.

### 4.4.1 Low voltage business single rate tariff

The low voltage business single rate tariff is available for non-residential low voltage customers with a type 1-6 meter installation. As with the residential tariff, consumption is charged on an inclining scale in four consumption blocks. The low voltage business single rate tariff incorporates the charging parameters set out in Table 6.

**Table 6 - Low voltage business single rate tariff charging parameters**

| Charging Parameter | Units  | Element of service  |                            |                              |                                                     |
|--------------------|--------|---------------------|----------------------------|------------------------------|-----------------------------------------------------|
|                    |        | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                         |
| Supply Rate        | \$/day | ✓                   | -                          | If applicable                | Fixed daily charge                                  |
| Block 1 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For consumption up to and including 833.3 kWh/month |
| Block 2 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the next 7,500 kWh/month                        |
| Block 3 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the next 8333.3 kWh/month                       |
| Block 4 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the balance of consumption                      |

A negotiated service charge applies where ETSA Utilities is required to read the Type 1-4 meter (eg. for tier one customers and for tier two customers < 160MWh pa).

A negotiated charge also applies for the monthly reading of Type 5-6 meters. This tariff is invoiced monthly or quarterly.

## 4.4.2 Low voltage business 2-rate tariff

The low voltage business 2-rate tariff has a Time of Use (ToU) structure with peak and off-peak consumption charges, using a type 1-6 meter. The customer's peak period energy consumption is charged in four consumption blocks. The low voltage business 2-rate tariff incorporates the charging parameters set out in Table 7.

**Table 7 - Low voltage business 2-rate tariff charging parameters**

| Charging Parameter      | Units  | Element of service  |                            |                              |                                                     |
|-------------------------|--------|---------------------|----------------------------|------------------------------|-----------------------------------------------------|
|                         |        | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                         |
| Supply Rate             | \$/day | ✓                   | -                          | If applicable                | Fixed daily charge                                  |
| Peak Block 1 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For consumption up to and including 833.3 kWh/month |
| Peak Block 2 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the next 7,500 kWh/month                        |
| Peak Block 3 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the next 8333.3 kWh/month                       |
| Peak Block 4 Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For the balance of consumption                      |
| Off Peak Usage Rate     | \$/kWh | ✓                   | ✓                          | -                            | For all off-peak period consumption                 |

Off Peak consumption is charged at a flat rate. A negotiated service charge applies where ETSA Utilities is required to read the type 1-4 meter or for the monthly reading of Type 5-6 meters. This tariff is invoiced monthly or quarterly.

## 4.4.3 Low voltage kVA demand tariff

The low voltage kVA demand tariff generally applies to business customers taking supply directly from a distribution transformer. This tariff requires a Type 5 (interval) meter capable of measuring both active and reactive power. A minimum demand of 80 kVA applies and a stepped kVA demand charge applies.

As with the business 2-rate tariff, the energy consumption of these customers has ToU peak and off-peak consumption charges. The low voltage kVA demand tariff incorporates the charging parameters set out in Table 8.

**Table 8 - Low voltage kVA demand tariff charging parameters**

| Charging Parameter         | Units      | Element of service  |                            |                              |                                                             |
|----------------------------|------------|---------------------|----------------------------|------------------------------|-------------------------------------------------------------|
|                            |            | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                                 |
| Supply Rate                | \$/day     | ✓                   | -                          | If applicable                | Fixed daily charge                                          |
| Annual Block 1 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For peak demand up to and including 100 kVA                 |
| Annual Block 2 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the next 150 kVA                                        |
| Annual Block 3 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the next 750 kVA                                        |
| Annual Block 4 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the balance of kVA demand                               |
| Additional demand          | \$/kVA/mth | ✓                   | -                          | -                            | Demand in excess of that required during summer peak hours. |
| Peak Usage Rate            | \$/kWh     | ✓                   | ✓                          | -                            | For all peak period energy consumption                      |
| Off Peak Usage Rate        | \$/kWh     | ✓                   | ✓                          | -                            | For all off-peak period energy consumption                  |

A negotiated service charge applies for the monthly reading of Type 5 meters.

These tariffs are invoiced monthly.

#### 4.4.4 Unmetered supply tariffs

Unmetered supply tariffs are applicable to supply points that are not metered. Energy consumption is calculated using the appropriate algorithm in the applicable Metrology Procedure. Unmetered tariffs comprise of an energy rate that is applied to the calculated electricity consumption.

There are two unmetered supply tariffs that are assigned by ETSA Utilities, depending upon the customer's consumption profile:

- Unmetered Overnight Usage supply is for overnight use for public lighting.
- Unmetered 24 Hour Usage supply is for constant 24 hour per day use, typically public phones, traffic lights and telecommunications installations.

The low voltage unmetered usage tariffs incorporate the charging parameter set out in Table 9.

**Table 9 - Unmetered overnight and 24 hour usage tariff charging parameters**

| Charging Parameter | Units  | Element of service  |                            |                              |                                  |
|--------------------|--------|---------------------|----------------------------|------------------------------|----------------------------------|
|                    |        | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                      |
| Anytime Usage Rate | \$/kWh | ✓                   | ✓                          | -                            | For estimated energy consumption |

Unmetered supply tariffs are generally invoiced monthly.

## 4.5 High Voltage Business Tariff Class

There is also a broad range of customer sizes and types connected to ETSA Utilities system at high voltage. They are predominantly manufacturing and commercial installations.

### 4.5.1 High voltage business 2-rate tariff

This business 2-rate tariff for customers connected at high voltage is similar in all respects, including the charging parameters, to the low voltage equivalent described in section 4.4.2.

### 4.5.2 High Voltage kVA demand tariff

The kVA demand tariff for business customers connected at high voltage is similar in structure to the low voltage equivalent described in section 4.4.3.

Two variants of this tariff are available with different charging parameters, depending upon whether the customer's maximum demand is in excess of 1000 kVA. Minimum demands of 80 kVA and 1,000 kVA apply to the respective tariffs. The charging parameters of these tariffs are set out in Table 10 and Table 11.

**Table 10 - High voltage kVA demand tariff charging parameters <1,000 kVA**

| Charging Parameter         | Units      | Element of service  |                            |                              |                                                             |
|----------------------------|------------|---------------------|----------------------------|------------------------------|-------------------------------------------------------------|
|                            |            | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                                 |
| Supply Rate                | \$/day     | ✓                   | -                          | If applicable                | Fixed daily charge                                          |
| Annual Block 1 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For peak demand up to and including 100 kVA                 |
| Annual Block 2 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the next 150 kVA                                        |
| Annual Block 3 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the next 750 kVA                                        |
| Annual Block 4 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the balance of kVA demand                               |
| Additional demand          | \$/kVA/mth | ✓                   | -                          | -                            | Demand in excess of that required during summer peak hours. |
| Peak Usage Rate            | \$/kWh     | ✓                   | ✓                          | -                            | For all peak period energy consumption                      |
| Off Peak Usage Rate        | \$/kWh     | ✓                   | ✓                          | -                            | For all off-peak period energy consumption                  |

**Table 11 - High voltage kVA demand tariff charging parameters**

| Charging Parameter         | Units      | Element of service  |                            |                              |                                                             |
|----------------------------|------------|---------------------|----------------------------|------------------------------|-------------------------------------------------------------|
|                            |            | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                                 |
| Supply Rate                | \$/day     | ✓                   | -                          | If applicable                | Fixed daily charge                                          |
| Annual Block 1 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For peak demand up to and including 1,000 kVA               |
| Annual Block 2 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the next 2,000 kVA                                      |
| Annual Block 3 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For the balance of kVA demand                               |
| Additional demand          | \$/kVA/mth | ✓                   | -                          | -                            | Demand in excess of that required during summer peak hours. |
| Peak Usage Rate            | \$/kWh     | ✓                   | ✓                          | -                            | For all peak period energy consumption                      |
| Off Peak Usage Rate        | \$/kWh     | ✓                   | ✓                          | -                            | For all off-peak period energy consumption                  |

## 4.6 Major Business Tariff Class

The major business customers are the largest connected to ETSA Utilities network. They comprise a range of industrial, manufacturing and mining enterprises.

### 4.6.1 High voltage zone substation kVA demand tariff

This kVA demand tariff is for larger high voltage connected business customers that take supply on direct distributors from a zone substation. The tariff has a minimum chargeable demand of 5,000 kVA and a minimum annual usage of 25 GWh. A type 1-4 interval meter is required with the ability to measure both active and reactive power. The charging parameters for this tariff are set out in Table 12.

**Table 12 - High voltage zone substation kVA demand tariff charging parameters**

| Charging Parameter         | Units      | Element of service  |                            |                              |                                                             |
|----------------------------|------------|---------------------|----------------------------|------------------------------|-------------------------------------------------------------|
|                            |            | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                                 |
| Supply Rate                | \$/day     | ✓                   | -                          | If applicable                | Fixed daily charge                                          |
| Annual Block 3 Demand Rate | \$/kVA/mth | ✓                   | ✓                          | -                            | For peak demand > 5,000 kVA                                 |
| Additional demand          | \$/kVA/mth | ✓                   | -                          | -                            | Demand in excess of that required during summer peak hours. |
| Peak Usage Rate            | \$/kWh     | ✓                   | ✓                          | -                            | For all peak period energy consumption > 25 GWh p.a.        |
| Off Peak Usage Rate        | \$/kWh     | ✓                   | ✓                          | -                            | For all off-peak period energy consumption                  |

A negotiated service charge may apply for the monthly reading of Type 1-4 meters and the tariffs is invoiced monthly.

### 4.6.2 High voltage zone substation kVA demand locational tariff

This kVA demand tariff is similar in structure to the high voltage zone substation kVA demand tariff described in section 4.6.1. It applies to those customers having a demand in excess of 10 MW or annual consumption greater than 40 GWh. A type 1-4 interval meter is required with the ability to measure both active and reactive power.

The tariff has averaged distribution charges, but a locational TUoS component is applied. The charging parameters of this tariff are shown in Table 13.

**Table 13 - High voltage zone substation kVA demand locational tariff charging parameters**

| Charging Parameter  | Units      | Element of service  |                            |                              |                                                             |
|---------------------|------------|---------------------|----------------------------|------------------------------|-------------------------------------------------------------|
|                     |            | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                                 |
| Supply Rate         | \$/day     | -                   | Locational                 | If applicable                | Fixed daily charge                                          |
| Annual Demand Rate  | \$/kVA/mth | ✓                   | -                          | -                            | For peak demand > 5,000 kVA                                 |
| Additional demand   | \$/kVA/mth | ✓                   | -                          | -                            | Demand in excess of that required during summer peak hours. |
| Peak Usage Rate     | \$/kWh     | ✓                   | -                          | -                            | For all peak period energy consumption > 25 GWh p.a.        |
| Off Peak Usage Rate | \$/kWh     | ✓                   | -                          | -                            | For all off-peak period energy consumption                  |

A negotiated service charge may apply for the monthly reading of Type 1-4 meters. This tariff is invoiced monthly.

## 4.6.3 Subtransmission kVA demand locational tariff

Subtransmission customers are connected to ETSA Utilities' network at 66 or 33 kV and have a minimum demand of 10 MW. The tariff for these customers is similar in structure to the high voltage zone substation kVA demand locational tariff and also has a locational TUoS component. The charging parameters for the tariff are set out in Table 14.

**Table 14 - Subtransmission kVA demand locational tariff charging parameters**

| Charging Parameter  | Units      | Element of service  |                            |                              |                                                             |
|---------------------|------------|---------------------|----------------------------|------------------------------|-------------------------------------------------------------|
|                     |            | Direct control DUoS | Transmission recovery TUoS | Alternative control Metering | Description                                                 |
| Supply Rate         | \$/day     | -                   | Locational                 | If applicable                | Fixed daily charge                                          |
| Annual Demand Rate  | \$/kVA/mth | ✓                   | -                          | -                            | For peak demand > 10,000 kVA                                |
| Additional demand   | \$/kVA/mth | ✓                   | -                          | -                            | Demand in excess of that required during summer peak hours. |
| Peak Usage Rate     | \$/kWh     | ✓                   | -                          | -                            | For all peak period energy consumption                      |
| Off Peak Usage Rate | \$/kWh     | ✓                   | -                          | -                            | For all off-peak period energy consumption                  |

A negotiated service charge may also apply, for the monthly reading of Type 1-4 meters and the tariffs is invoiced monthly.

## 5 Network tariff strategy

This section contains the objectives that ETSA Utilities applies to the development of its network tariffs. It goes on to outline the strategies ETSA Utilities proposes to pursue in developing tariffs during the 2010-15 regulatory control period.

### 5.1 Regulatory Requirements

The information in this section concerning potential future network tariff developments is provided pursuant to clause 6.18.9 of the Rules.

#### 6.18.9 Publication of information about tariffs and tariff classes

- (a) A *Distribution Network Service Provider* must maintain on its website:
- (3) a statement of expected price trends (to be updated for each *regulatory year*) giving an indication of how the *Distribution Network Service Provider* expects prices to change over the *regulatory control period* and the reasons for the expected changes.

### 5.2 Network tariff objectives

This section presents the high level framework that ETSA Utilities applies to the development of its network tariff strategy. The major objectives of network pricing are to some extent conflicting and therefore involve making compromises. They are as follows:

- **Revenue sufficiency** - prices are formulated to recover permitted weighted average prices under the determination.
- **Revenue volatility** - to the extent possible, tariffs will be structured to minimise monthly and annual variations in revenue.
- **Pricing efficiency** - through their variable components, prices will signal the economic cost of providing network service. Residual costs will be recovered in a manner which least distorts customers' consumption decisions.
- **Customer equity** - customers should pay a reasonable allocated share of costs and moves towards pricing cost reflectivity need to be tempered to limit their impact on some customers.
- **Pricing stability** - to the extent possible undue variation in price levels should be avoided.
- **Pricing simplicity** - price structures should be understandable, simple and transparent.

### 5.3 The need for tariff reform

As explained in section 3.5, to a greater extent than any other Australian distributor, ETSA Utilities' summer demand is sensitive to the effect of air conditioning demand. High summer peak demands occur during heat wave conditions, which correspond with periods when the elements of the system have least capacity and the power factor of loads is poor.

Some 75% of the capital expenditure on ETSA Utilities' network in the 2010-15 regulatory control period is growth related. That is, the expenditure is driven by the need to augment and expand the network to adequately meet peak summer demand and provide for the connection of new customers.

As a consequence, the management of summer demand has a high priority in ETSA Utilities' tariff reform strategies. This leads to an emphasis on providing network price signals that will encourage both residential and business customers to manage their demand by the following means:

- The price levels of existing tariff structures;
- The development of more cost reflective tariff structures; and
- The development of innovative new tariff structures.

## **5.4 Network tariff strategy**

ETSA Utilities has a pricing strategy that will, within the limitations of metering arrangements and efficient tariff structures, signal the costs associated with increased demand placed on the network, including the use of air conditioning, within.

Consistent with the network tariff objectives outlined in section 5.2, ETSA Utilities' network tariff strategy aims to:

- Attain revenue sufficiency under the Weighted Average Price Cap;
- Signal the long run marginal cost of supply through its network tariffs;
- Improve cost reflectivity and reduce revenue variability by reducing the reliance on usage based tariff components where appropriate;
- Pass on the cost of Electranet's transmission services to customers; and
- Explore tariff based demand management opportunities, including voluntary time of use tariffs, and tariffs that target network constraints on a locational basis. Sections 5.6 and 5.8 outline future tariff reform options under consideration and development.

## **5.5 Tariff reform 2005 to 2010**

During the 2005-10 regulatory control period, ETSA Utilities undertook a number of tariff reforms, including the following:

- The restructuring of the single rate tariffs for residential and low voltage business tariffs to introduce a greater number of steps and progressively increasing consumption charges for the higher consumption blocks;
- Businesses have been encouraged to adopt kVA demand price structures; and
- Power factor correction for businesses has been facilitated through the use an excluded/negotiated service charge for excess reactive power requirements.

## 5.6 Experimental tariff programs

Over the past few years, ETSA Utilities Network has launched a number of pilot programs to better understand customer behaviour, facilitate improved pricing outcomes and development of new tariffs. The following are examples of innovative tariffs used in Australia and abroad that are currently being considered for a pilot within ETSA Utilities Network's area:

- **Capacity Pricing** – the use of a capacity price rather than energy, small customers;
- **Direct load control** – where the DNSP can control appliances at the customer's premises such as air conditioning to moderate the demand on the network; and
- **Voluntary load control** – where the customer may be offered a discount or rebate in return for responding to the DNSP's request to reduce their demand.

ETSA Utilities Network is constantly monitoring innovative tariffs around the world for ideas that may be suited for the South Australian market. Promising new ideas will therefore be incorporated in the Pricing Strategy on an ongoing basis.

## 5.7 ETSA Utilities' demand management and intelligent network trials

ETSA Utilities is about to embark on the final stage of a large-scale demand management project in the suburb of North Adelaide in 2011/12, involving approximately 5,000 customers. The objective of this project is to further develop the direct load control programs pioneered in Glenelg and Mawson Lakes since 2006. These techniques will be coupled with a range of intelligent network solutions, using:

- Network sensing devices;
- Advanced metering infrastructure;
- Wireless 4G communications and Home Area Network communications; and
- Developing the associated back-office information systems.

Intelligent technologies and customer equipment to be trialled in the North Adelaide area and potentially, deployed to other areas include:

- Electric vehicles, their charging infrastructure and control methods;
- Energy storage and control methods;
- Air conditioning demand management control interfaces; and
- Distribution equipment automation.

This program will involve the recruitment of customers to participate in direct load control and other trials late in 2011. Whilst the associated incentives that will be trialled with these customers fall outside the standard control services in this submission, the learnings from the trial may well influence the future development of tariffs.

## **5.8 Future tariff reform options**

ETSA Utilities proposes to explore any benefits capable of being realised from continuing the network tariff reform process. Within the limitations of the AER's DMIA provisions, ETSA Utilities may obtain more first-hand information on customers usage of the network through trials and associated customer research.

The following network tariff reforms may be pursued by ETSA Utilities during the remainder of the 2010-15 regulatory control period:

- Improving the design of the ToU and demand tariffs, to enhance their cost reflectivity;
- Strengthening the signal of the single-rate inclining block structure as a 'second best' option to ToU and demand pricing, where interval meters are not available;
- Continuing to assign more customers to network demand tariffs where it is economically efficient to do so; and
- Exploring tariff based demand management opportunities.

In relation to the reassignment of customers to individual tariffs within the tariff classes, ETSA Utilities believes that it is appropriate to actively encourage and facilitate the development of more cost reflective distribution network tariffs. Such tariffs may well include those that are enabled by the use of developing technologies, such as:

- Communications signalling and load management technologies, such as that already employed by ETSA Utilities on a trial basis for its Peakbreaker+ program<sup>14</sup>; or
- Remotely read interval meters that are not the subject of a regulatory obligation or requirement.

For cost reflective tariffs and technologies to be employed, a flexible approach to the reassignment of customers to tariffs is required. The facilitation of tariff related demand management would align with the AER's established direction with non-tariff based demand management.

ETSA Utilities envisages that it will be seeking to implement the following tariff initiatives during the remainder of the 2010-15 determination:

- Increasingly, interval meters are being chosen as an alternate meter for replacements and new connections. This would permit the introduction of more cost reflective capacity based prices for some small customers, including some residential customers.
- In specific situations where network demand management is sought, ETSA Utilities may offer voluntary tariffs from time to time and, should the customer or their retailer accept this offer, they would be reassigned to the new tariff.

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<sup>14</sup> ETSA Utilities Peakbreaker+ program is a refinement of earlier direct load control schemes. A communications module is fitted alongside the customer's conventional electricity meter and is capable of controlling air conditioning compressors on a rotational basis. It has two way radio communications, which offers a range of additional features to its load switching capability.

These proposed initiatives are intended to improve the cost reflectivity of the existing tariffs and whilst involving the reassignment of customers to new tariffs, would not require their reassignment to a new or different tariff class.

## **5.9 Expected DUoS price trends 2010 - 2015**

For tariffs in place at the commencement of the 2010-15 regulatory control period, ETSA Utilities' tariff strategy and its focus on managing demand leads to the indicative movement in tariff charging parameters summarised in Table 15. The actual price movements each year will remain subject to review at the time, following consideration of the objectives set out in section 5.2 of this Pricing Proposal.

**Table 15 - Indicative relative charging parameter movement in the 2010-15 regulatory control period**

| Distribution tariff class and tariff | Fixed charge | First block rate | Upper block rates | Peak energy rate | Off peak energy rate | Peak demand rate | Off peak demand rate |
|--------------------------------------|--------------|------------------|-------------------|------------------|----------------------|------------------|----------------------|
| Residential                          |              |                  |                   |                  |                      |                  |                      |
| Residential flat rate                | ↑            | —                | ↑                 |                  |                      |                  |                      |
| Controlled load                      |              | —                | ↑                 |                  |                      |                  |                      |
| Low voltage business                 |              |                  |                   |                  |                      |                  |                      |
| LV business flat rate                | ↑            | —                | ↑                 |                  |                      |                  |                      |
| LV business 2—rate                   | ↑            |                  | ↑                 | —                | —                    |                  |                      |
| LV business kVA demand               | ↑            |                  |                   | —                | —                    | —                | —                    |
| High Voltage business                |              |                  |                   |                  |                      |                  |                      |
| HV business kVA demand               | ↑            |                  |                   | —                | —                    | —                | —                    |
| Major business                       |              |                  |                   |                  |                      |                  |                      |
| ST business kVA demand               |              |                  |                   | —                | —                    | —                | —                    |

↑ increase relative to the average distribution price movement permitted in the AER's 2010 Decision

↓ decrease relative to the average distribution price movement permitted in the AER's 2010 Decision

— no material change relative to the average distribution price movement permitted in the AER's 2010 Decision

A blank cell indicates that the corresponding charging parameter is not applicable for a particular tariff.

## 6 Standard control services tariffs

Within the framework of ETSA Utilities' longer term tariff strategy set out in chapter 5 of this Pricing Proposal, this section sets out the proposed rates for tariff charging components of standard control services for 2011/12 and provides a comparison with the rates in place during 2010/11.

It should be noted that the information and comparisons in this section relate solely to standard control services. ETSA Utilities' distribution charges to customers are bundled charges that contain transmission cost recovery components and may also contain alternative control metering service components.

A discussion of customer impacts including the transmission cost recovery and alternative service metering components is set out in chapter 7 of this Pricing Proposal.

### 6.1 Regulatory Requirements

The information in this section concerning the change in standard control service rates is provided pursuant to clause 6.18.9 of the Rules.

#### 6.18.9 Publication of information about tariffs and tariff classes

- (a) A *Distribution Network Service Provider* must maintain on its website:
  - (3) a statement of expected price trends (to be updated for each *regulatory year*) giving an indication of how the *Distribution Network Service Provider* expects prices to change over the *regulatory control period* and the reasons for the expected changes.

### 6.2 Tariff changes in 2011/12

The tariff changes in 2011/12 have been made within the overall tariff strategy for the 2010-15 regulatory control period set out in section 5. The opportunity for progressive rebalancing of tariff components has been curtailed by the overall magnitude of the price change this year. The distribution component of tariffs increased by a weighted average of 29.77% and the transmission related component by 20.7%.

To avoid passing through price changes that would be significantly larger than the average, the changes to the distribution tariffs in 2011/12 therefore comprised:

- An increase in the small customer fixed charge, to the extent permitted by the side constraint; and
- A increase in the energy rates of the residential tariff to provide an overall 29.7% increase. The rate for the first energy block was increased by 39.7% and the upper energy block rates by 30.5%;
- A uniform percentage increase of the demand and energy rates of the business tariffs by 30.5%, to provide an overall 29.8% increase.

ETSA Utilities proposes to continue with its tariff rebalancing objectives, to achieve greater pricing cost reflectivity, in 2012/13.

## 6.3 Calculation of network use of system tariffs

The following sections provide information on the DUoS component of ETSA Utilities' existing tariffs and the price movement proposed for 2011/12. It should be noted that this information is provided for the purpose of showing the relative change in the price of each tariff charging parameter. Compliance with clause 6.18.2(8) of the Rules, concerning the demonstration that price changes comply with the Rules and the AER's Determination, is demonstrated in Section 8 and Appendix G of this Pricing Proposal.

All prices in the following sections are exclusive of GST and have been rounded.

## 6.4 Low Voltage Residential tariff class

### 6.4.1 Low voltage residential tariff

Table 16 contains the proposed 2011/12 prices for single rate residential customers.

**Table 16 - Proposed 2011/12 low voltage residential tariff single rate**

| Charging Parameter | Units  | Existing 2010/11 | Proposed 2011/12 | Variance |
|--------------------|--------|------------------|------------------|----------|
| Supply Rate        | \$p.a. | 81.09            | 86.67            | 5.57     |
| Block 1 Usage Rate | ¢/kWh  | 5.417            | 7.567            | 2.149    |
| Block 2 Usage Rate | ¢/kWh  | 7.622            | 9.947            | 2.325    |
| Block 3 Usage Rate | ¢/kWh  | 9.068            | 11.834           | 2.766    |
| Block 4 Usage Rate | ¢/kWh  | 9.068            | 11.834           | 2.766    |

### 6.4.2 Controlled load tariff

Table 17 contains the proposed 2011/12 prices for controlled load customers.

**Table 17 - Proposed 2011/12 controlled load tariff**

| Charging Parameter | Units | Existing 2010/11 | Proposed 2011/12 | Variance |
|--------------------|-------|------------------|------------------|----------|
| Block 1 Usage Rate | ¢/kWh | 2.226            | 2.905            | 0.679    |
| Block 2 Usage Rate | ¢/kWh | 3.023            | 3.945            | 0.460    |

## 6.5 Low Voltage Business tariff class

### 6.5.1 Low voltage business single rate tariff

Table 18 contains the proposed 2011/12 prices for low voltage business tariff single customers.

**Table 18 - Proposed 2011/12 low voltage business single rate tariff**

| Charging Parameter | Units  | Existing 2010/11 | Proposed 2011/12 | Variance |
|--------------------|--------|------------------|------------------|----------|
| Supply Rate        | \$p.a. | 81.09            | 86.67            | 5.57     |
| Block 1 Usage Rate | c/kWh  | 6.555            | 8.555            | 1.999    |
| Block 2 Usage Rate | c/kWh  | 7.741            | 10.102           | 2.361    |
| Block 3 Usage Rate | c/kWh  | 8.202            | 10.703           | 2.502    |
| Block 4 Usage Rate | c/kWh  | 8.202            | 10.703           | 2.502    |

### 6.5.2 Low voltage business 2-rate tariff

Table 19 contains the proposed 2011/12 prices for low voltage business 2-rate tariff customers.

**Table 19 - Proposed 2011/12 low voltage business 2-rate tariff**

| Charging Parameter      | Units  | Existing 2010/11 | Proposed 2011/12 | Variance |
|-------------------------|--------|------------------|------------------|----------|
| Supply Rate             | \$p.a. | 81.09            | 86.67            | 5.57     |
| Peak Block 1 Usage Rate | c/kWh  | 8.624            | 11.254           | 2.630    |
| Peak Block 2 Usage Rate | c/kWh  | 8.624            | 11.254           | 2.630    |
| Peak Block 3 Usage Rate | c/kWh  | 8.624            | 11.254           | 2.630    |
| Peak Block 4 Usage Rate | c/kWh  | 8.624            | 11.254           | 2.630    |
| Off Peak Usage Rate     | c/kWh  | 2.723            | 3.553            | 0.831    |

### 6.5.3 Low voltage business kVA demand tariff

Table 20 contains the proposed 2011/12 prices low kVA demand business tariff customers.

**Table 20 - Proposed 2011/12 low voltage business kVA demand tariff**

| Charging Parameter         | Units    | Existing 2010/11 | Proposed 2011/12 | Variance |
|----------------------------|----------|------------------|------------------|----------|
| Supply Rate                | \$p.a.   | 0.00             | 0.00             | 0.00     |
| Annual Block 1 Demand Rate | \$/kVA/m | 8.285            | 10.812           | 2.527    |
| Annual Block 2 Demand Rate | \$/kVA/m | 4.833            | 6.307            | 1.474    |
| Annual Block 3 Demand Rate | \$/kVA/m | 3.539            | 4.618            | 1.079    |
| Annual Block 4 Demand Rate | \$/kVA/m | 2.675            | 3.491            | 0.816    |
| Additional Demand          | \$/kVA/m | 2.675            | 3.491            | 0.816    |
| Peak Usage Rate            | c/kWh    | 1.709            | 2.230            | 0.521    |
| Off Peak Usage Rate        | c/kWh    | 1.709            | 2.230            | 0.521    |

#### 6.5.4 Unmetered supply tariffs

Proposed 2011/12 prices for the unmetered supply tariff customers are set out in Table 21 and Table 22.

**Table 21 - Proposed 2011/12 Unmetered Overnight tariff**

| Charging Parameter | Units | Existing 2010/11 | Proposed 2011/12 | Variance |
|--------------------|-------|------------------|------------------|----------|
| Anytime Usage Rate | ¢/kWh | 3.754            | 4.899            | 1.145    |

**Table 22 - Proposed 2011/12 Unmetered 24 hour Usage tariff**

| Charging Parameter | Units | Existing 2010/11 | Proposed 2011/12 | Variance |
|--------------------|-------|------------------|------------------|----------|
| Anytime Usage Rate | ¢/kWh | 3.340            | 4.358            | 1.019    |

### 6.6 High Voltage Business tariff class

#### 6.6.1 High voltage business 2-rate tariff

The proposed 2011/12 prices for high voltage business 2-rate customers are the same as those for the low voltage equivalent tariff in Table 19.

#### 6.6.2 High Voltage kVA demand tariff

Table 23 contains the proposed 2011/12 prices for High Voltage kVA demand business tariff customers.

**Table 23 - Proposed 2011/12 high voltage business kVA demand tariff**

| Charging Parameter         | Units    | Existing 2010/11 | Proposed 2011/12 | Variance |
|----------------------------|----------|------------------|------------------|----------|
| Supply Rate                | \$p.a.   | 0.00             | 0.00             | 0.00     |
| Annual Block 1 Demand Rate | \$/kVA/m | 3.406            | 4.445            | 1.039    |
| Annual Block 2 Demand Rate | \$/kVA/m | 2.063            | 2.692            | 0.629    |
| Annual Block 3 Demand Rate | \$/kVA/m | 1.588            | 2.072            | 0.484    |
| Additional Demand          | \$/kVA/m | 1.588            | 2.072            | 0.484    |
| Peak Usage Rate            | ¢/kWh    | 1.473            | 1.922            | 0.449    |
| Off-Peak Usage Rate        | ¢/kWh    | 1.473            | 1.922            | 0.449    |

## 6.7 Major Business tariff class

### 6.7.1 High voltage zone substation kVA demand tariff

Table 24 contains the proposed 2011/12 prices for zone substation kVA demand business customers.

**Table 24 - Proposed 2011/12 zone substation kVA demand business tariff**

| Charging Parameter  | Units    | Existing 2010/11 | Proposed 2011/12 | Variance |
|---------------------|----------|------------------|------------------|----------|
| Supply Rate         | \$p.a.   | 0.00             | 0.00             | 0.00     |
| Annual Demand Rate  | \$/kVA/m | 1.588            | 2.072            | 0.484    |
| Additional Demand   | \$/kVA/m | 1.105            | 1.442            | 0.337    |
| Peak Usage Rate     | c/kWh    | 0.518            | 0.676            | 0.158    |
| Off Peak Usage Rate | c/kWh    | 0.518            | 0.676            | 0.158    |

### 6.7.2 High voltage zone substation kVA demand locational tariff

Table 25 contains the proposed 2011/12 prices for zone substation kVA demand locational tariff business customers.

**Table 25 - Proposed 2011/12 Zone substation kVA demand locational tariff**

| Charging Parameter  | Units    | Existing 2010/11 | Proposed 2011/12 | Variance |
|---------------------|----------|------------------|------------------|----------|
| Supply Rate         | \$p.a.   | 0.00             | 0.00             | 0.00     |
| Annual Demand Rate  | \$/kVA/m | 1.588            | 2.072            | 0.484    |
| Additional Demand   | \$/kVA/m | 1.105            | 1.442            | 0.337    |
| Peak Usage Rate     | c/kWh    | 0.518            | 0.676            | 0.158    |
| Off Peak Usage Rate | c/kWh    | 0.518            | 0.676            | 0.158    |

### 6.7.3 Subtransmission kVA demand locational tariff

Table 26 contains the proposed 2011/12 prices for subtransmission kVA demand locational tariff business customers.

**Table 26 - Proposed 2011/12 Subtransmission kVA demand locational tariff**

| Charging Parameter  | Units    | Existing 2010/11 | Proposed 2011/12 | Variance |
|---------------------|----------|------------------|------------------|----------|
| Supply Rate         | \$p.a.   | 0.00             | 0.00             | 0.00     |
| Annual Demand Rate  | \$/kVA/m | 0.484            | 0.631            | 0.148    |
| Additional Demand   | \$/kVA/m | 0.484            | 0.631            | 0.148    |
| Peak Usage Rate     | c/kWh    | 0.173            | 0.226            | 0.053    |
| Off Peak Usage Rate | c/kWh    | 0.173            | 0.226            | 0.053    |

## 7 Customer Impacts

In this chapter, customer impacts are calculated using ETSA Utilities' proposed bundled NUoS tariffs. The use of these network tariffs results in customer impacts that include the following components:

- DUoS charges, for ETSA Utilities' standard control services;
- Metering charges, for ETSA Utilities' alternative control services; and
- Transmission cost recovery tariffs, to recover costs associated with Electranet's standard control services and avoided TUoS payments made by ETSA Utilities to embedded generators.

All of the customer impacts presented in this chapter are GST exclusive.

This chapter provides an indication of how the price trends of tariff classes may be expected to change over the 2010-15 regulatory control period.

The price changes for individual tariffs have been calculated on the basis that the most recent available customer consumption volume levels, for the 12 months to March 2011. Where the number of customers on a tariff is sufficient, the distribution of price impacts has been presented.

This chapter also sets out how ETSA Utilities complies with the AER's requirement for a system of tariff review, where the charge varies according to the usage or load profile of a customer.

### 7.1 Regulatory Requirements

#### 7.1.1 Rules requirements

Rules clause 6.18.2 imposes a requirement for the Pricing Proposal to set out the nature of variations that may take place during 2011/12.

#### 6.18.2 Pricing proposals

(b) A *pricing proposal* must:

- (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the *regulatory year* and the basis on which it could occur;

Clause 6.18.9 requires the publication of price trends over the 2010-15 regulatory control period and the reasons for expected variations.

#### 6.18.9 Publication of information about tariffs and tariff classes

(a) A *Distribution Network Service Provider* must maintain on its website:

- (3) a statement of expected price trends (to be updated for each *regulatory year*) giving an indication of how the *Distribution Network Service Provider* expects prices to change over the *regulatory control period* and the reasons for the expected changes.

## **7.1.2 Requirements of the AER's Decision**

Appendix B of the AER's decision contains the following requirement based on clause 6.18.4(b) of the Rules, concerning the review of tariffs where the charge varies according to the usage or load profile of a customer<sup>15</sup>.

### **System of assessment and review of the basis on which a customer is charged**

11. Where the charging parameters for a particular tariff result in a basis of charge that varies according to the customer's usage or load profile, ETSA Utilities must set out in its pricing proposal a method of how it will review and assess the basis on which a customer is charged.
12. If the AER considers that the method provided under section 11 does not provide for an effective system of assessment and review of the basis on which a customer is charged, the AER may request additional information or request that ETSA Utilities revise and resubmit a revised method.
13. If the AER considers the method provided in accordance with section 11 is reasonable it will approve that method by notice in writing to ETSA Utilities.

## **7.2 Overall price trends during the 2010-15 regulatory control period**

The expected overall price trends over the course of the determination are built up from the following components:

- ETSA Utilities' distribution prices for standard control services, in accordance with the AER's 2010 determination, as amended by the decisions of the Australian Competition Tribunal in relation to the inclusion to easements in the regulatory asset base and the corporate taxation allowance described in section 2.3.3;
- ETSA Utilities' prices for alternative control metering services, in accordance with the AER's 2010 determination, as amended by the decision of the Australian Competition Tribunal in relation to the corporate taxation allowance described in section 2.3.4; and
- Transmission related prices, estimated from Electranet's revenue determination.

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<sup>15</sup> AER, Final Decision - South Australia distribution determination 2010–11 to 2014–15, May 2010, pp. 287.

These ‘bundled’ prices are projected in Table 27, for the 2010-15 regulatory control period.

**Table 27 – Network price trends over the regulatory period (nominal)**

| Tariff Class   | Tariff                  | Actual  |                | Forecast        |                 |                |                |
|----------------|-------------------------|---------|----------------|-----------------|-----------------|----------------|----------------|
|                |                         | 2009/10 | 2010/11        | 2011/12         | 2012/13         | 2013/14        | 2014/15        |
| LV Residential | Residential             | \$88.84 | \$96.67<br>9%  | \$123.92<br>28% | \$141.26<br>14% | \$152.47<br>8% | \$157.26<br>3% |
|                | Residential & Hot Water | \$65.06 | \$71.31<br>10% | \$91.38<br>28%  | \$104.16<br>14% | \$112.42<br>8% | \$115.95<br>3% |
| LV Business    | Business Single         | \$93.72 | \$98.76<br>5%  | \$124.22<br>26% | \$141.35<br>14% | \$152.28<br>8% | \$157.02<br>3% |
|                | Business 2-Rate         | \$67.22 | \$78.21<br>16% | \$99.60<br>27%  | \$113.30<br>14% | \$122.01<br>8% | \$125.80<br>3% |
|                | Business LV Demand      | \$50.11 | \$59.55<br>19% | \$76.29<br>28%  | \$84.27<br>10%  | \$90.57<br>7%  | \$93.36<br>3%  |
| HV Business    | Business HV Demand      | \$38.36 | \$41.14<br>7%  | \$52.43<br>27%  | \$57.46<br>10%  | \$61.41<br>7%  | \$63.25<br>3%  |
| Major Business | Zone 5/Std Locational   |         | \$19.19        | \$23.70<br>24%  | \$26.54<br>12%  | \$28.09<br>6%  | \$28.90<br>3%  |
|                | Sub-Trans Locational    |         | \$16.21        | \$19.53<br>20%  | \$21.55<br>10%  | \$22.44<br>4%  | \$23.03<br>3%  |

## 7.3 Variations to prices

Clause 6.18.2(b)(5) requires ETSA Utilities to set out the nature of any variations and adjustments that could occur to tariffs during the course of the 2011/12 year. In addition, clause 6.18.9(a)(3) requires an indication of how ETSA Utilities expects prices to change over the regulatory control period and the reasons for the expected changes.

Variations to the determination during the course of the 2010-15 regulatory control period could result in the adjustment of distribution network tariffs from the price trends set out in Table 27. Those variations that are reasonably foreseen would arise from the following effects:

- The offsetting effect of market settlements surpluses on TUoS charges;
- The service target performance incentive scheme (STPIS)
- The efficiency benefit sharing scheme (EBSS);
- The demand management incentive scheme (DMIS);
- Any cost adjustments in respect of undergrounding of powerlines as directed by the Minister;
- Variation from forecast of carryover revenue adjustments from the former regulatory period (termed EDPD amounts); and
- Approved pass through amounts.

The extent of these variations to price during the course of the 2010-15 regulatory control period will depend on a number of factors, including ETSA Utilities performance as measured against the parameters of the AER’s incentive schemes. However, the overall effect on prices is anticipated to be of the order of a few percent only.

With regard to network price variations that could occur during the 2011/12 year, network prices will be established in accordance with this Pricing Proposal for implementation on 1 July 2011. Those prices will not vary throughout 2011/12.

The average price trends mask the variation in price that can take place for individual customers. Each customer's price will vary depending upon their level of consumption, and for business customers, the load profile and monthly demand. The following information on expected price impacts relates to the likely range of price effects. These have been determined from applying the most recent full year of consumption data, to March 2011, to the 2011/12 prices.

## 7.4 Low Voltage Residential tariff class

### 7.4.1 Low voltage residential tariff

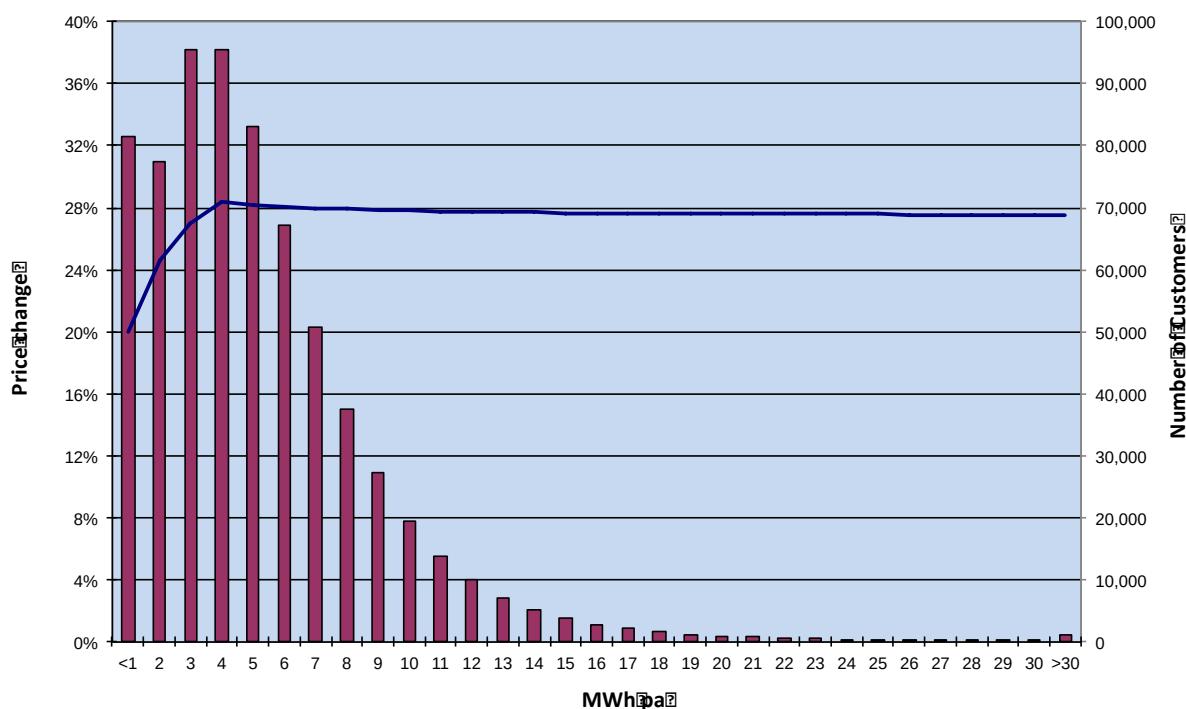
The low voltage residential tariff has a single-rate with an inclining block structure and four consumption steps. The 2011/12 annual bill and price change for this tariff is shown in Table 28, for a range of different customer consumption levels.

**Table 28 - Low voltage residential price change in 2011/12**

| Annual Usage | NUoS 2010/11 | NUoS 2011/12 | Change in NUoS bill | DUoS 2010/11 | DUoS 2011/12 | Change in DUoS bill |
|--------------|--------------|--------------|---------------------|--------------|--------------|---------------------|
| MWh pa       | \$ pa        | \$ pa        | %                   | \$ pa        | \$ pa        | %                   |
| 2            | 230          | 286          | 24.6%               | 189          | 238          | 25.7%               |
| 4            | 378          | 486          | 28.4%               | 298          | 389          | 30.8%               |
| 5            | 483          | 620          | 28.2%               | 374          | 489          | 30.7%               |
| 8            | 799          | 1,022        | 27.9%               | 603          | 787          | 30.6%               |
| 16           | 1,764        | 2,252        | 27.7%               | 1,299        | 1,696        | 30.6%               |

Figure 4 illustrates how the effect of the price change varies with customer's annual energy consumption and also shows the distribution of customers on the residential tariff.

**Figure 4 - Low voltage residential NUoS price change in 2011/12**



## 7.4.2 Controlled load tariff

The controlled load tariff also has an inclining block structure with two consumption steps. The 2011/12 annual bill and price change shown in Table 29 for residential customers with hot water, for a range of representative consumption levels.

**Table 29 - Low voltage residential + hot water price change in 2011/12**

| Annual Usage | NUoS 2010/11 | NUoS 2011/12 | Change in NUoS bill | DUoS 2010/11 | DUoS 2011/12 | Change in DUoS bill |
|--------------|--------------|--------------|---------------------|--------------|--------------|---------------------|
| MWh pa       | \$ pa        | \$ pa        | %                   | \$ pa        | \$ pa        | %                   |
| 2 + 1        | 259          | 323          | 25.0%               | 212          | 267          | 26.2%               |
| 4 + 2        | 436          | 560          | 28.3%               | 342          | 447          | 30.7%               |
| 5 + 3        | 570          | 731          | 28.1%               | 441          | 576          | 30.7%               |
| 8 + 4        | 915          | 1,170        | 27.9%               | 692          | 903          | 30.6%               |
| 16 + 5       | 1,910        | 2,438        | 27.7%               | 1,410        | 1,841        | 30.6%               |

## 7.5 Low Voltage Business tariff class

### 7.5.1 Low voltage business single rate tariff

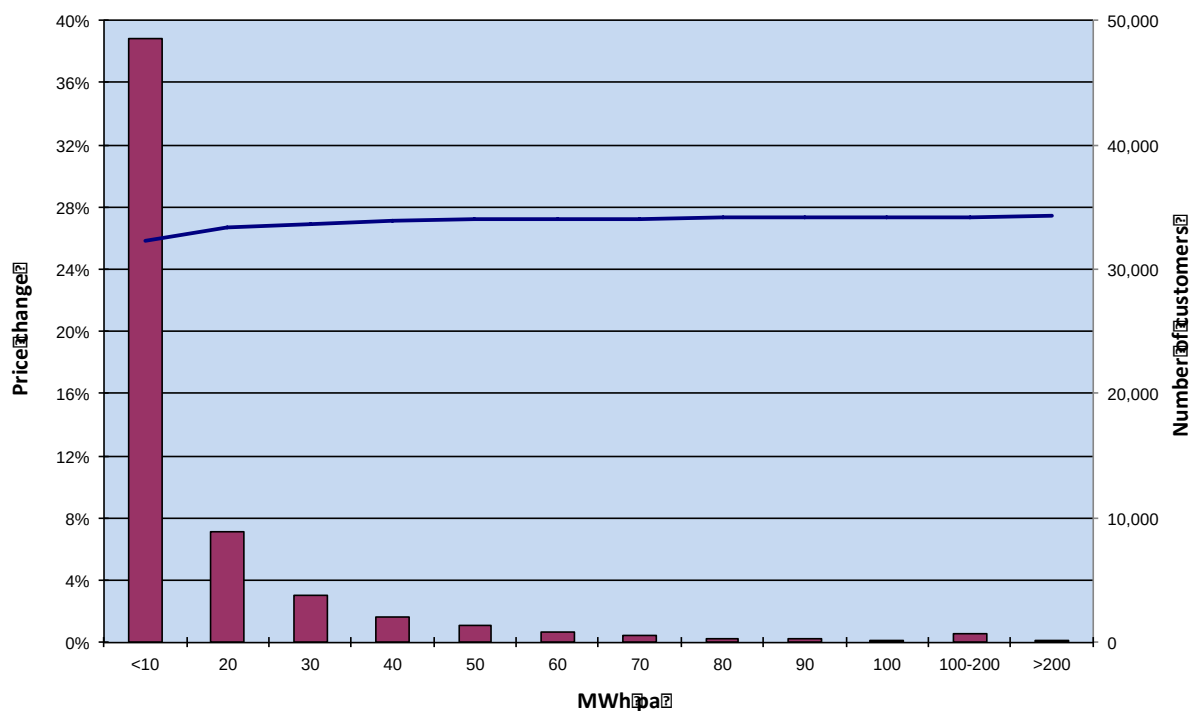
The low voltage business single rate tariff has an anytime consumption charge with an inclining block structure and four consumption steps. Table 30 shows the 2011/12 annual bill and price change for this tariff, for a range of annual consumption levels.

**Table 30 - Low voltage business single rate NUoS price change in 2011/12**

| Annual Usage<br>MWh pa | NUoS 2010/11<br>\$ pa | NUoS 2011/12<br>\$ pa | Change in NUoS bill<br>% | DUoS 2010/11<br>\$ pa | DUoS 2011/12<br>\$ pa | Change in DUoS bill<br>% |
|------------------------|-----------------------|-----------------------|--------------------------|-----------------------|-----------------------|--------------------------|
| 4                      | 444                   | 549                   | 23.7%                    | 343                   | 429                   | 24.9%                    |
| 10                     | 988                   | 1,242                 | 25.8%                    | 737                   | 942                   | 27.9%                    |
| 20                     | 2,053                 | 2,600                 | 26.7%                    | 1,511                 | 1,952                 | 29.2%                    |
| 50                     | 5,248                 | 6,674                 | 27.2%                    | 3,833                 | 4,983                 | 30.0%                    |
| 150                    | 16,252                | 20,702                | 27.4%                    | 11,804                | 15,386                | 30.3%                    |

The price change versus annual consumption and the distribution of customers for the low voltage business single rate tariff is illustrated in Figure 5.

**Figure 5 - Low voltage business single rate NUoS price change in 2011/12**



## 7.5.2 Low voltage business 2-rate tariff

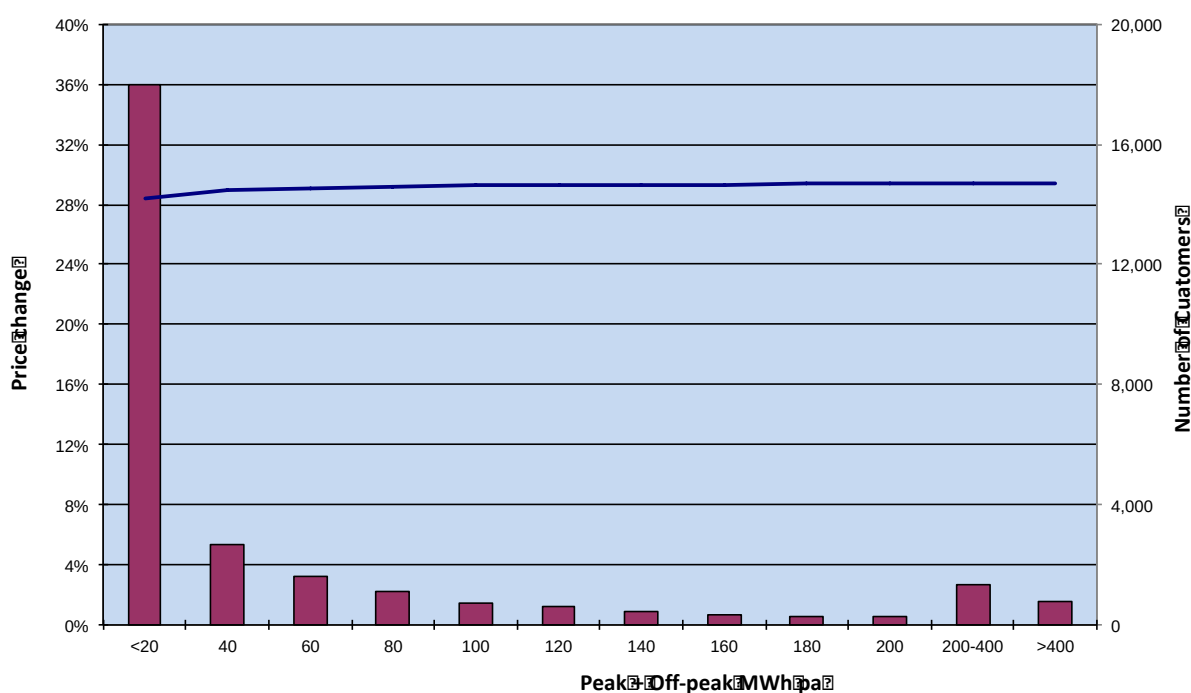
The effect of the price change in 2011/12 for more complex tariff structures such as the low voltage business 2-rate will depend upon the customer consumption profile, in particular the ratio of peak to off-peak period usage. Table 31 shows how the 2011/12 annual bill has changed for this tariff, for different customer consumption levels and average peak to off peak consumption proportions of 50%.

**Table 31 - Low voltage business 2-rate NUoS price change in 2011/12**

| Annual Usage<br>MWh pa | NUoS 2010/11<br>\$ pa | NUoS 2011/12<br>\$ pa | Change in NUoS bill<br>% | DUoS 2010/11<br>\$ pa | DUoS 2011/12<br>\$ pa | Change in DUoS bill<br>% |
|------------------------|-----------------------|-----------------------|--------------------------|-----------------------|-----------------------|--------------------------|
| 8                      | 700                   | 877                   | 25.2%                    | 535                   | 679                   | 26.9%                    |
| 20                     | 1,629                 | 2,061                 | 26.5%                    | 1,216                 | 1,567                 | 28.9%                    |
| 40                     | 3,177                 | 4,036                 | 27.0%                    | 2,350                 | 3,048                 | 29.7%                    |
| 100                    | 7,821                 | 9,960                 | 27.4%                    | 5,755                 | 7,491                 | 30.2%                    |
| 300                    | 23,299                | 29,707                | 27.5%                    | 17,101                | 22,298                | 30.4%                    |

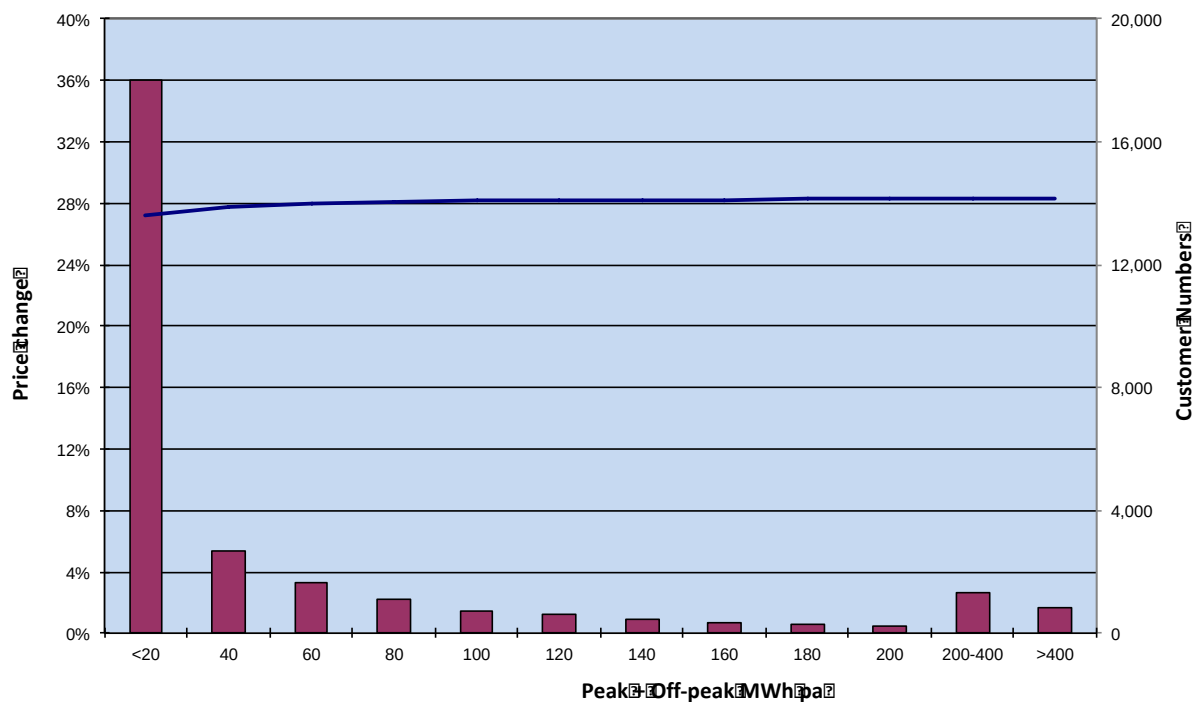
A bar chart displaying the bill impact vs customer consumption is shown in Figure 6 and Figure 7. A small number of high voltage business 2-rate customers on their identical tariff have been included in this illustration.

**Figure 6 - Low voltage business 2-rate NUoS change in 2011/12, 50% peak consumption**



The higher prices that would apply to business 2 rate customers with a peak to off peak consumption proportion of 70% is illustrated in the bar chart in Figure 7.

**Figure 7 - Low voltage business 2-rate NUoS change in 2011/12, 70% peak consumption**

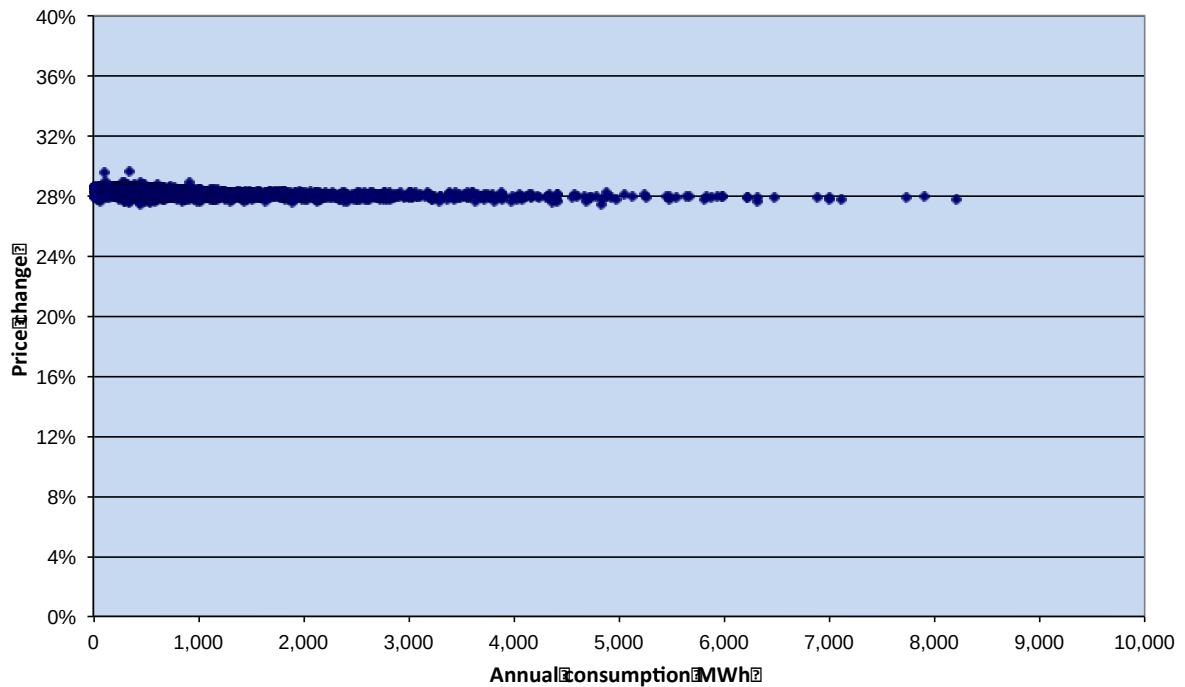


### 7.5.3 Low Voltage kVA demand tariff

The average annual price change of low voltage kVA demand customers in 2011/12 is 24.6%. However, there is significant variation in the price change due to the differing load and power factors of these customers.

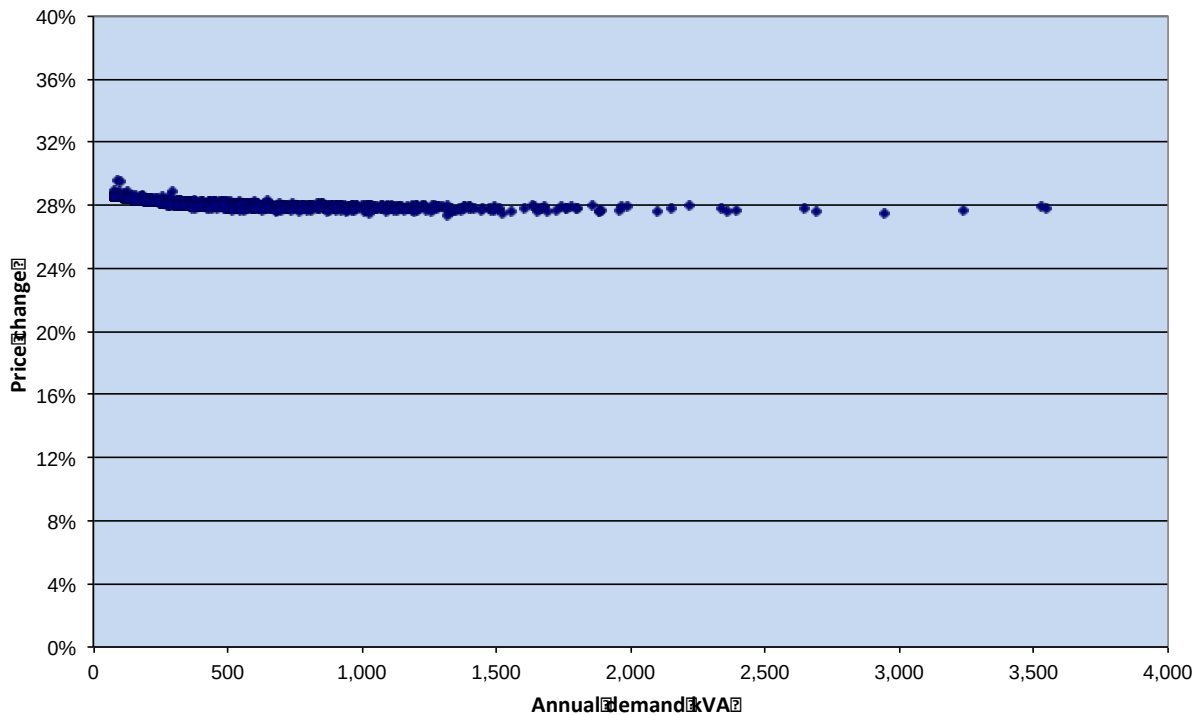
A scatter chart displaying the bill impact vs customer annual energy consumption in MWh is shown in Figure 8.

**Figure 8 - Low voltage business kVA NUoS change vs consumption in 2011/12**



A second scatter chart displaying the bill impact vs customer annual demand in kVA is shown in Figure 9.

**Figure 9 - Low voltage business kVA NUoS change vs demand in 2011/12**



## 7.5.4 Unmetered supply tariffs

The average 2011/12 price and price change from 2010/11 for the two unmetered supply tariffs is shown in Table 32.

**Table 32 - Unmetered supply NUoS 2011/12 average price and price change**

| Tariff        | NUoS              |                   |             | DUoS              |                   |             |
|---------------|-------------------|-------------------|-------------|-------------------|-------------------|-------------|
|               | 2010/11<br>\$/MWh | 2011/12<br>\$/MWh | change<br>% | 2010/11<br>\$/MWh | 2011/12<br>\$/MWh | change<br>% |
| Unmetered 12h | 49.74             | 63.99             | 28.6%       | 37.54             | 48.99             | 30.5%       |
| Unmetered 24h | 43.81             | 56.37             | 28.7%       | 33.40             | 43.58             | 30.5%       |

## 7.6 High Voltage Business tariff class

### 7.6.1 High voltage business 2-rate tariff

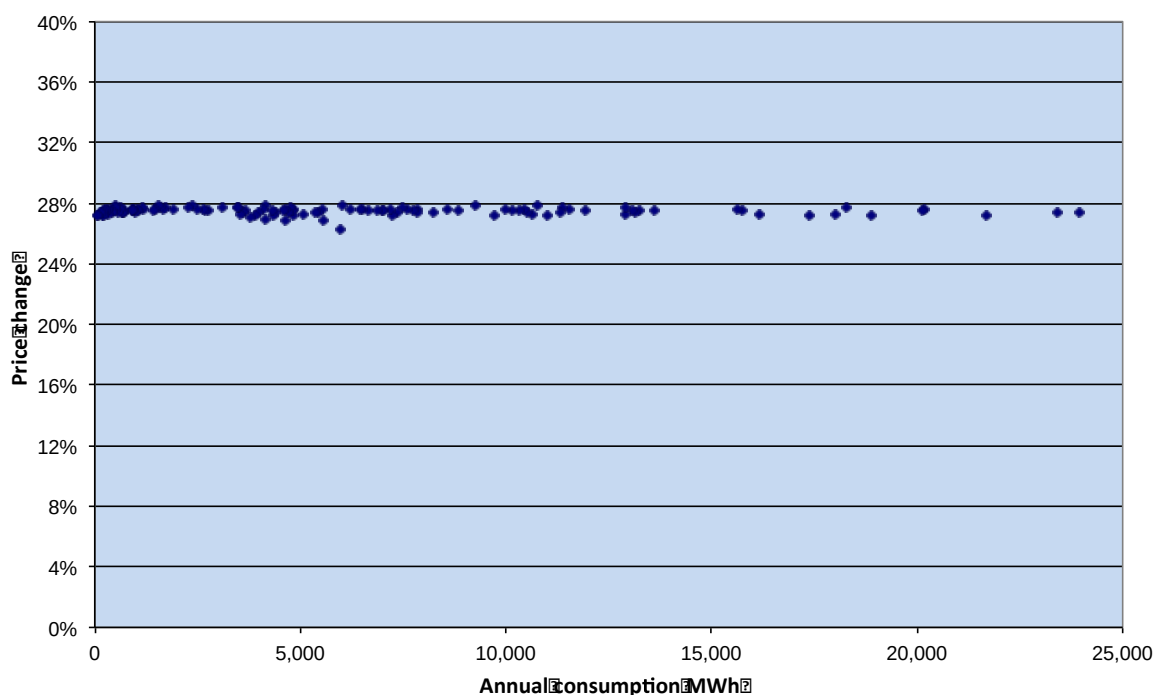
The rates of the high voltage and low voltage business 2-rate tariffs are the same. The small number of High Voltage 2-rate customers were included in Figure 6 for the purpose of displaying the effect of the 2011/12 price change.

### 7.6.2 High Voltage kVA demand tariff

The average annual price change of high voltage kVA demand customers in 2011/12 is 25.2%. However, as with the low voltage kVA customers, there is significant variation due to the differing load and power factors of these customers.

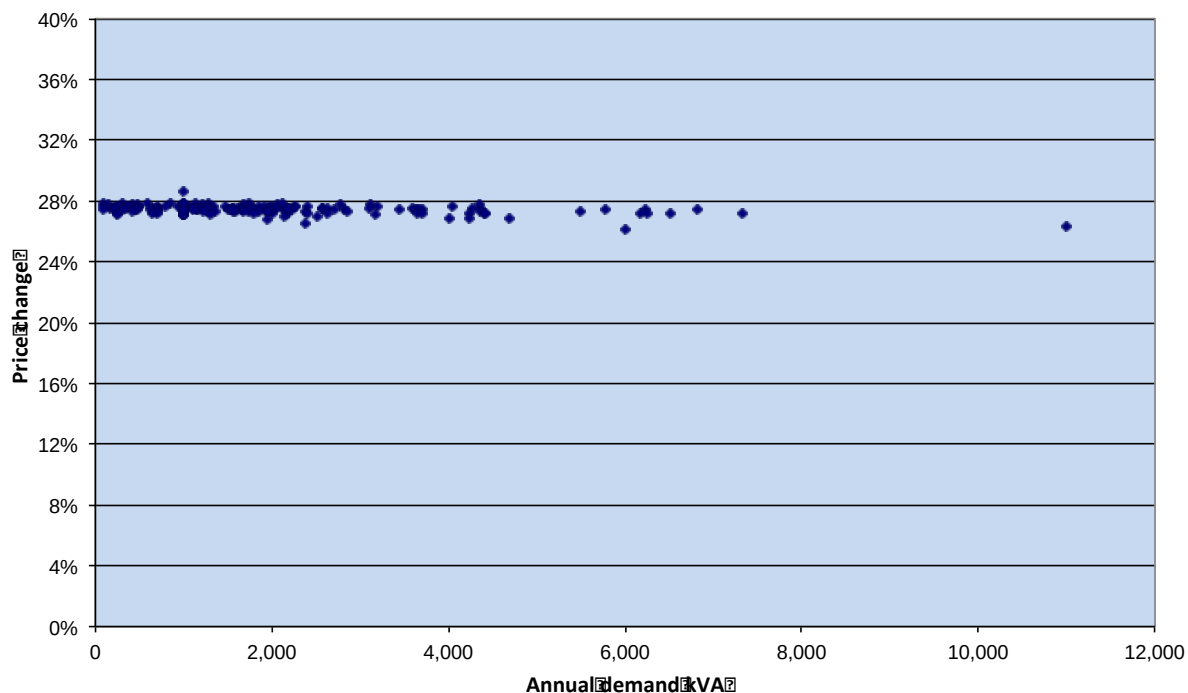
A scatter chart displaying the bill impact vs customer annual energy consumption in MWh is shown in Figure 10.

**Figure 10 - High voltage business kVA NUoS change vs annual consumption in 2011/12**



A second scatter chart displaying the bill impact vs customer annual demand in kVA is shown in Figure 11.

**Figure 11 - High voltage business kVA NUoS change vs annual demand in 2011/12**



## 7.7 Major Business tariff class

### 7.7.1 High voltage zone substation and Subtransmission kVA demand locational tariffs

There are a relatively small number of major business customers with locational TUoS charges and accordingly these are not amenable to a scatter chart illustration of the 2011/12 price change.

There is little variability between the individual price changes for these customers and the averages for the tariff. The average NUoS price change for these customers in 2011/12 is illustrated in Table 27 and is 23% for high voltage zone substation customers and 24% for subtransmission customers.

## 7.8 Review of customer charges

Pursuant to clause 6.18.4(b) of the Rules, the AER has set out the requirement for a system of assessment and review of the basis on which a customer is charged where the charge varies with the customer's usage or profile. This requirement is in Appendix B, clause 11 of the Decision.

Sections 7.4 to 7.7 of this Pricing Proposal illustrate the price changes which result from ETSA Utilities' proposed prices, in 2011/12. These sections contain illustrations that demonstrate not just the average price change, but highlight the expected distribution of price changes, for customers with a range of consumption profiles and tariffs, over the full range of charging parameters.

In establishing the distribution prices in the 2010/11 Pricing Proposal, ETSA Utilities gave close consideration to the range of price changes and in particular to 'outlying' customers, where variation in the components of their particular consumption profile could give rise to a price change which differs markedly from the average for the tariff and tariff class. However, in this 2011/12 Pricing proposal, the increase in tariff price components has been uniformly applied to demand and energy. As a result, customers generally do not need to be reassigned to different tariffs in 2011/12. It is considered that the transfers required from the annual review in 2011/12 are not material for the purpose of this Pricing Proposal

In the 2010/11 Pricing Proposal, ETSA Utilities established this approach to the assessment and review of the basis on which customers are charged. The following principal factors are considered in this review, which is conducted annually at the time that tariffs are established and tariff strategies are reviewed:

- The consumption levels of each customer in a tariff are reviewed, to determine those customers for which the tariff may be inappropriate; and
- The price impact on individual customers also forms the basis for a review of the suitability of the basis of charging.

As a result of this annual review, ETSA Utilities would contact any relevant customer that was assessed as being charged on an unsuitable basis, to propose either an alternative network tariff or measures by which the customer's network charge might be efficiently reduced. This approach will be continued throughout the 2010-15 regulatory control period.

The outcome of this process is illustrated by three on-going tariff programs, as examples of how ETSA Utilities has proactively approached customers, with a view to modifying the basis on which the customers are charged:

- ETSA Utilities is currently encouraging larger single and two-rate low voltage business customers to transfer to the low voltage kVA demand tariff. The inclining and stepped energy rates of these energy based tariffs result in relatively high average charges for larger customers, for which the tariffs were not designed. Customers that transfer to the more cost reflective kVA demand tariff have a greater range of opportunities to respond to the pricing signals;
- In a similar manner, during the past five years ETSA Utilities has now largely eliminated what was an extensive range of obsolete kW demand network tariffs, by encouraging the affected customers to transfer to the most appropriate current tariff; and
- Since 2007, ETSA Utilities has been approaching business customers on kVA demand tariffs that have non-compliant (poor) power factors, to provide information on how those customers may economically improve their power factor and reduce network charges.

ETSA Utilities considers that this process complies with the requirement of clause 11 of Appendix B of the AER's Decision in formulating the 2011/12 network prices. Moreover, by providing detailed information on the distribution of network price changes in this and

subsequent Pricing Proposals, the overall effectiveness of this tariff review process may be assessed.

## 8 Pricing of standard control services

This section demonstrates how ETSA Utilities' network tariffs for 2011/12 comply with the requirements of the Rules and the AER's determination in respect of the pricing X factors, side constraints and pricing principles.

In 2010, ETSA Utilities engaged KPMG as an independent consultant to review this Pricing Proposal for the compliance with the relevant sections of the Rules and the AER's Determination.

### 8.1 Regulatory requirements

#### 8.1.1 Rules requirements

Rules clause 6.18.2(b) specifies that ETSA Utilities' Pricing Proposal must contain information concerning the expected revenue to be derived from its tariff classes and tariffs, as follows.

#### 6.18.2 Pricing proposals

(b) A *pricing proposal* must:

- (4) set out, for each *tariff class* related to *standard control services*, the expected weighted average revenue for the relevant *regulatory year* and also for the current *regulatory year*; and

In setting its prices for standard control services, clause 6.18.5 of the Rules requires ETSA Utilities to comply with the following pricing principles.

#### 6.18.5 Pricing principles

- (a) For each *tariff class*, the revenue expected to be recovered should lie on or between:
  - (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and
  - (2) a lower bound representing the avoidable cost of not serving those customers.
- (b) A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:
  - (1) must take into account the long run marginal cost for the service or, in the case of a *charging parameter*, for the element of the service to which the *charging parameter* relates; and
  - (2) must be determined having regard to:
    - (i) transaction costs associated with the tariff or each *charging parameter*; and
    - (ii) whether customers of the relevant *tariff class* are able or likely to respond to price signals.
- (c) If, however, as a result of the operation of paragraph (b), the *Distribution Network Service Provider* may not recover the expected revenue, the provider must adjust its

tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

In respect of pricing side constraints, ETSA Utilities is required to comply with clause 6.18.6 of the Rules:

## 6.18.6 Side constraints on tariffs for standard control services

- (a) This clause applies only to *tariff classes* related to the provision of *standard control services*.
- (b) The expected weighted average revenue to be raised from a *tariff class* for a particular *regulatory year* of a *regulatory control period* must not exceed the corresponding expected weighted average revenue for the preceding *regulatory year* by more than the permissible percentage.
- (c) The permissible percentage is the greater of the following:
  - (1) the CPI-X limitation on any increase in the *Distribution Network Service Provider's* expected weighted average revenue between the two *regulatory years* plus 2%;  
*Note:*  
*The calculation is of the form  $(1 + \text{CPI})(1 - X)(1 + 2\%)$*
  - (2) CPI plus 2%.  
*Note:*  
*The calculation is of the form  $(1 + \text{CPI})(1 + 2\%)$*
- (d) In deciding whether the permissible percentage has been exceeded in a particular *regulatory year*, the following are to be disregarded:
  - (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13;
  - (2) the recovery of revenue to accommodate pass through of charges for *transmission use of system services* to customers.
- (e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.

ETSA Utilities' tariffs also subject to South Australian transitional provision Clause 9.29.5(d) of the Rules, which limits the movement of the fixed supply charge component for small customers (with a consumption less than 160 MWh per annum).

## 8.1.2 Requirements of the AER's Decision

The principal elements of the AER's determination are set out in its Decision and form the major determinants of prices for standard control services during the 2010-15 regulatory control period:

Chapter 4 - Pricing control mechanism.

Side constraint requirements

Chapter 16 - Pricing X factors

These elements of the AER's Decision have been set out in section 2.3 of this Pricing Proposal.

In addition, the provisions of Appendix E to the AER's Decision, concerning changes to tariff structures, must be met.

## 8.2 2011/12 prices for standard control services

The fundamental pricing criteria that ETSA Utilities has factored into this Pricing Proposal are summarised in Table 33<sup>16</sup>.

**Table 33 - Summary of fundamental pricing criteria**

| Criterion                                         | 2011/12 value |
|---------------------------------------------------|---------------|
| Consumer Price Index                              | 3.33%         |
| X Factor                                          | -18.10%       |
| S Factor                                          | 0.00%         |
| D Factor                                          | 0.00%         |
| U Factor                                          | 0.00%         |
| EDPD Factor                                       | 6.34%         |
| WAPC $(1+CPI)x(1-X)x(1+S)x(1+U)x(1+D)x(1+EDPD)-1$ | 29.77%        |

The derivation of the WAPC constraint is presented in the AER's template provided for this purpose (refer Appendix G of this Proposal). Derivation of the EDPD factor is detailed in Appendix C and the derivation of the pass through amount is set out in Appendix J of this Proposal. The remaining adjustment terms of the WAPC equation are zero in 2011/12 for the following reasons:

- The carryover adjustment from the service incentive scheme under the 2005-10 EDPD is included within the EDPD adjustment. Consequently, the S factor will not commence until audited data for 2010/11 is available, in 2012/13;
- Expenditure on demand management initiatives to require a D factor adjustment will not commence until audited data is available, in 2012/13; and
- The undergrounding adjustment U has not been triggered by any South Australian government requirement for undergrounding during 2010/11.

<sup>16</sup> AER, Final Decision, May 2010, p. 251.

## 8.3 Compliance with the Weighted Average Price Cap

The AER's WAPC model has been used for the purposes of demonstrating compliance with the provisions of the WAPC. This model is submitted as Appendix G (confidential) and forms part of this pricing Proposal.

The prices and side constraints for 2011/12 are based on 2010/11 volumes, projected using the WAPC formulae and X factors determined by the AER.

A small number of customers on obsolete kW demand tariffs in 2009/10 (year *t-2*) were transferred to kVA tariffs before the end of that financial year. The historical data therefore needs to be modified to permit the obsolete tariff to be removed from the WAPC in 2010/11 and 2011/12. The provisions of Appendix E of the AER's decision were therefore followed in developing reasonable estimates of volumes for tariff charging parameters to which the customers were transferred.

These tariff charging parameter changes and the associated reasonable estimates of consumption are detailed in Appendix I (Confidential) of this Pricing Proposal. The use of reasonable estimates in 2011/12 was minimal.

A summary of the tariff class revenue is presented in Table 34.

**Table 34 – Weighted Average Revenue<sup>17</sup>**

| Tariff class   | 2010/11<br>Revenue<br>\$'000 | 2011/12<br>Revenue<br>\$'000 | Change in<br>weighted<br>average<br>revenue<br>% | WAPC change<br>in weighted<br>average<br>revenue<br>% |
|----------------|------------------------------|------------------------------|--------------------------------------------------|-------------------------------------------------------|
| Major business | 8,357                        | 10,905                       | 30.50%                                           |                                                       |
| HV Business    | 24,845                       | 32,423                       | 30.50%                                           |                                                       |
| LV Business    | 250,001                      | 324,302                      | 29.72%                                           |                                                       |
| LV Residential | 321,375                      | 416,930                      | 29.73%                                           |                                                       |
| Total          | 604,578                      | 784,560                      | 29.77%                                           | 29.77%                                                |

Table 38 demonstrates that ETSA Utilities' 2011/12 network Pricing Proposal complies with the WAPC X factor constraint indicated in Table 33 above. The table also satisfies clause 6.18.2(b)(4) of the Rules.

<sup>17</sup> Utilising *t-2* quantities, as required by the WAPC formula.

## 8.4 Tariff class side constraints

### 8.4.1 Tariff class movement side constraint

The side constraint formula that the AER has determined for ETSA Utilities has been set out in section 2.3.2 of this Pricing Proposal. The evaluation of the side constraint for 2011/12 is set out in Table 35 and Table 36<sup>18</sup>.

**Table 35 - Summary of side constraint criteria**

| Criterion                                                                                                 | 2011/12 value |
|-----------------------------------------------------------------------------------------------------------|---------------|
| Consumer Price Index                                                                                      | 3.33%         |
| X Factor                                                                                                  | -18.10%       |
| S Factor                                                                                                  | 0.00%         |
| D Factor                                                                                                  | 0.00%         |
| U Factor                                                                                                  | 0.00%         |
| EDPD Factor                                                                                               | 6.34%         |
| Side constraint $(1+CPI) \times (1-X) \times (1+S) \times (1+U) \times (1+D) \times (1+EDPD) (1+2\%) - 1$ | 32.36%        |

The AER's model has been used for the purposes of demonstrating compliance with the provisions of the side constraint. A summary of the tariff class revenue and price changes is presented in Table 36.

**Table 36 – Compliance with the side constraint**

| Tariff class   | 2010/11 Revenue<br>\$'000 | 2011/12 Revenue<br>\$'000 | Change in weighted<br>average revenue<br>% | Side<br>constraint<br>% |
|----------------|---------------------------|---------------------------|--------------------------------------------|-------------------------|
| Major Business | 8,357                     | 10,905                    | 30.50%                                     | 32.36%                  |
| HV Business    | 24,845                    | 32,423                    | 30.50%                                     | 32.36%                  |
| LV Business    | 250,001                   | 324,302                   | 29.72%                                     | 32.36%                  |
| LV Residential | 321,375                   | 416,930                   | 29.73%                                     | 32.36%                  |

Table 36 demonstrates ETSA Utilities' compliance with the provisions of clause 6.18.6 of the Rules and the AER's side constraint formula.

### 8.4.2 Fixed supply charge side constraint for small customers

In the transitional Rules, Clause 9.29.5(d) limits the maximum increase in the fixed supply charge component for small customers to \$10 per annum. ETSA Utilities has prepared tariffs for both standard control services and alternative control services that comply with this constraint. The alternative control services tariffs are described in section 11.

The fixed supply charge component of the tariff has two components: the DUoS Supply rate; and the alternative control metering charge for Type 5 meters. The increase in price of these components has been limited to \$5.57 and \$4.43, respectively.

<sup>18</sup> AER, Final Decision, May 2010, p. 251.

The fixed DUoS charges for small customers are illustrated in Table 37.

**Table 37 – Compliance with the fixed charge side constraint for small customers**

| Tariff Class   | 2010/11 Fixed Charge |          | 2011/12 Fixed Charge |          | Difference | Side constraint |
|----------------|----------------------|----------|----------------------|----------|------------|-----------------|
|                | ¢/day                | \$p.a.   | ¢/day                | \$p.a.   |            |                 |
| LV Residential |                      |          |                      |          |            |                 |
| DUoS           | 22.218               | \$81.09  | 23.745               | \$86.67  | \$5.57     |                 |
| Metering       | 5.850                | \$21.35  | 7.062                | \$25.78  | \$4.43     |                 |
| <b>Total</b>   | 28.068               | \$102.45 | 30.807               | \$112.45 | \$10.00    | \$10.00         |
| LV Business    |                      |          |                      |          |            |                 |
| DUoS           | 22.218               | \$81.09  | 23.745               | \$86.67  | \$5.57     |                 |
| Metering       | 5.850                | \$21.35  | 7.062                | \$25.78  | \$4.43     |                 |
| <b>Total</b>   | 22.218               | \$81.09  | 23.745               | \$86.67  | \$5.57     | \$10.00         |

Table 37 demonstrates ETSA Utilities' compliance with the provisions of clause 6.18.6 of the Rules and the AER's side constraint formula.

## 8.5 Compliance with pricing principles

This section demonstrates ETSA Utilities' compliance with the pricing principles set out in clause 6.18.5 of the Rules, which requires ETSA Utilities to ensure that the revenue recovered for each tariff class lies between:

- An upper bound, representing the stand-alone cost of serving the customers who belong to that class; and
- A lower bound, representing the avoidable cost of not serving those customers.

The Stand-alone and Avoidable cost methodologies are described in detail in Appendix F of this Proposal. These approaches are used to calculate the revenues for each standard control services tariff class associated with each cost methodology. These costs are compared with the weighted average revenue derived from ETSA Utilities' proposed tariffs.

### 8.5.1 Definition of Stand-alone and Avoidable costs

These two categories of cost may be defined for tariff classes, as follows:

- The **Stand-alone cost** for a tariff class is the cost of supplying only the tariff class concerned, with all other tariff classes not being supplied. If customers were to pay above the stand-alone cost then it would be economically beneficial for customers to switch to an alternative provider, and economically feasible for an alternative provider to operate. This creates the possibility of inefficient bypass of the existing infrastructure; and
- The **Avoidable cost** for a tariff class is the reduction in network cost that would take place if the tariff class were not supplied (whilst all other tariff classes remained supplied). If customers were to be charged below the avoidable cost, it would be economically beneficial for the business to stop supplying the customers and as the associated costs would exceed the revenue obtained from the customer.

There are two alternative concepts that could be used to calculate these costs:

- To ignore the sunk nature of the existing network and estimate the costs which would be associated with an optimally designed network, constructed to supply standard control services to the tariff class(es) concerned; or
- To base the estimation of costs on the modification of the existing network to provide standard control services to the tariff class(es) concerned

The Rules do not provide prescribe the methodology that should be used to calculate the Stand-alone and Avoidable costs of tariff classes of the network. ETSA Utilities has chosen to base its cost estimations on the second concept, with hypothetical modification of the existing network, rather than by devising and costing optimal new network structures. This has been done for two reasons:

- To avoid the very substantial resource requirements that would be involved in a full network redesign; and
- In recognition that the economic regulatory framework for distribution supports the existence and value of existing (sunk) network investments and does not support the optimisation of existing networks.

ETSA Utilities' approach is thus conceptually the same as that which was employed by Integral Energy in its 2009 Pricing Proposal, and subsequently approved by the AER<sup>19</sup>.

The derivation of the stand-alone and avoidable cost for tariff classes uses a methodology based on ETSA Utilities' cost of supply model. This model has been in use for a decade and was formulated to permit the efficient allocation of network costs to tariff classes and the formulation of network prices. The cost allocation model has been adapted to include the allocation of non system costs to tariff classes.

This model is described in the following sections. This approach was independently reviewed by KPMG, and certified to be compliant with the Rules in 2010. The approach has not been altered from that used in the 2010/11 Pricing Proposal and the sole change has been to index the outcomes by CPI in 2011.

## 8.6 Stand-alone costs

The stand-alone cost for each tariff class was derived from an engineering estimate of the proportions of the cost of providing network capacity that would need to remain in place to service the load in each of the four tariff classes in turn if the other three tariff classes were no longer required to be supplied.

The resulting cost allocation for each tariff class has been expressed in \$/kVA, using the estimated coincident contribution to ETSA Utilities' peak system demand and after adjustment for the average power factor of the tariff class.

The detailed procedure used to calculate the stand-alone cost of tariff classes is set out in Appendix F of this Pricing Proposal.

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<sup>19</sup> = Integral Energy, Direct Control Services - Initial Pricing Proposal 2009/10, 25 May 2009, p. 66-67.

## 8.7 Avoidable costs

In similar manner to the stand-alone cost, the avoidable cost associated with each of the four tariff classes were derived from an engineering estimate made of the network cost that could be avoided, in the event that each of the four tariff classes were to be removed in turn.

As with the stand-alone cost, the avoidable network cost was expressed in \$/kVA for each tariff class. The procedure used to calculate the avoidable cost of tariff classes is also set out in Appendix F of this Pricing Proposal.

## 8.8 Compliance with Rules clause 6.18.5(a)

The revenue expected to be recovered from each of ETSA Utilities' tariff classes in 2011/12 is compared with the stand-alone and avoidable costs calculated in sections 8.6 and 8.7 in Table 38.

**Table 38 - Stand-alone and avoidable distribution network costs (\$ M)**

| Tariff class   | Stand-alone cost | Tariff revenue | Avoidable cost |
|----------------|------------------|----------------|----------------|
| Major business | \$74             | \$12           | \$4            |
| HV business    | \$83             | \$33           | \$3            |
| LV business    | \$380            | \$315          | \$70           |
| LV residential | \$516            | \$388          | \$205          |

ETSA Utilities' network tariff classes lie within the subsidy free range, in that the expected DUoS revenue collected from the tariff class lies between the avoidable and stand alone costs of supply and therefore meet the requirements of section 6.18.5(a) of the Rules.

## 8.9 Long Run Marginal Costs

In the context of an infrastructure business with assets lives of typically 40 years, short run costs are effectively zero up to the point where the capacity of the network is exceeded. It is therefore important to use the Long Run Marginal Cost (LRMC) in any consideration of the marginal cost of providing network service.

There are three general approaches to the calculation of LRMC. Marsden Jacob Associates articulated these alternatives in its review of possible approaches, for the Queensland Competition Authority. This review was undertaken in the context of determining efficient prices for the Gladstone Area water Board. The alternatives are as follows<sup>20</sup>:

- **Marginal Incremental Cost (MIC)** - where a scenario involving increased demand is tested for its incremental effect on capex. The associated marginal cost is calculated as the difference between the present value of the investment programs divided by the increment in demand;

<sup>20</sup> Marsden Jacob Associates, Estimation of Long Run Marginal Cost (LRMC) - A report prepared by the Queensland Competition Authority - Final, 3 November 2004

- **Average Incremental Cost (AIC)** - is the present value of the incremental investment associated with increasing demand divided by the present value of the increment in demand; and
- **Long Run Incremental Cost (LRIC)** - is the annuitised value of the capital expenditure divided by the increment in demand.

In comparing the suitability of these three approaches, the first requires scenario definition and analysis, which although not impracticable would be very resource intensive in the context of a distribution business. Marsden Jacob make the point that because the AIC approach is based on a long term planning period, it takes a longer view of costs and provides more stable prices than the LRIC approach<sup>21</sup>.

The approach used by ETSA Utilities in this Pricing Proposal to determine the LRM of its tariff classes may be characterised as the AIC. However, the calculation has been extended to incorporate the incremental operating cost associated with new capital investment. This is the same as the approach taken by EnergyAustralia in its 2009 Pricing Proposal<sup>22</sup>.

ETSA Utilities' approach to the calculation of the network LRM for its tariff classes is set out in Appendix E of this Proposal. The approach has not been altered from that used in the 2010/11 Pricing Proposal and the sole change has been to index the outcomes by CPI in 2011.

These calculations are carried out at the following voltage levels of the network:

- Subtransmission;
- High voltage; and
- Low voltage.

The marginal cost at each network voltage level has been determined using the following relationship:

$$LRMC(AIC) = \frac{PV(growth\ related\ capex) + PV(growth\ related\ opex)}{PV(Incremental\ demand)}$$

Where:

*growth related capex* is the annualised capital expenditure to meet the additional demand and new customer connections forecast over the forecast period;

*growth related opex* is the incremental annual cost of operating and maintaining the newly constructed network and connection assets over the forecast period; and

*incremental demand* is the forecast change in kVA demand compared with the base year of 2009/10

The LRM outcomes at the subtransmission and high voltage levels are directly applicable to the Major Business and High Voltage Business tariff classes. At low voltage, a small

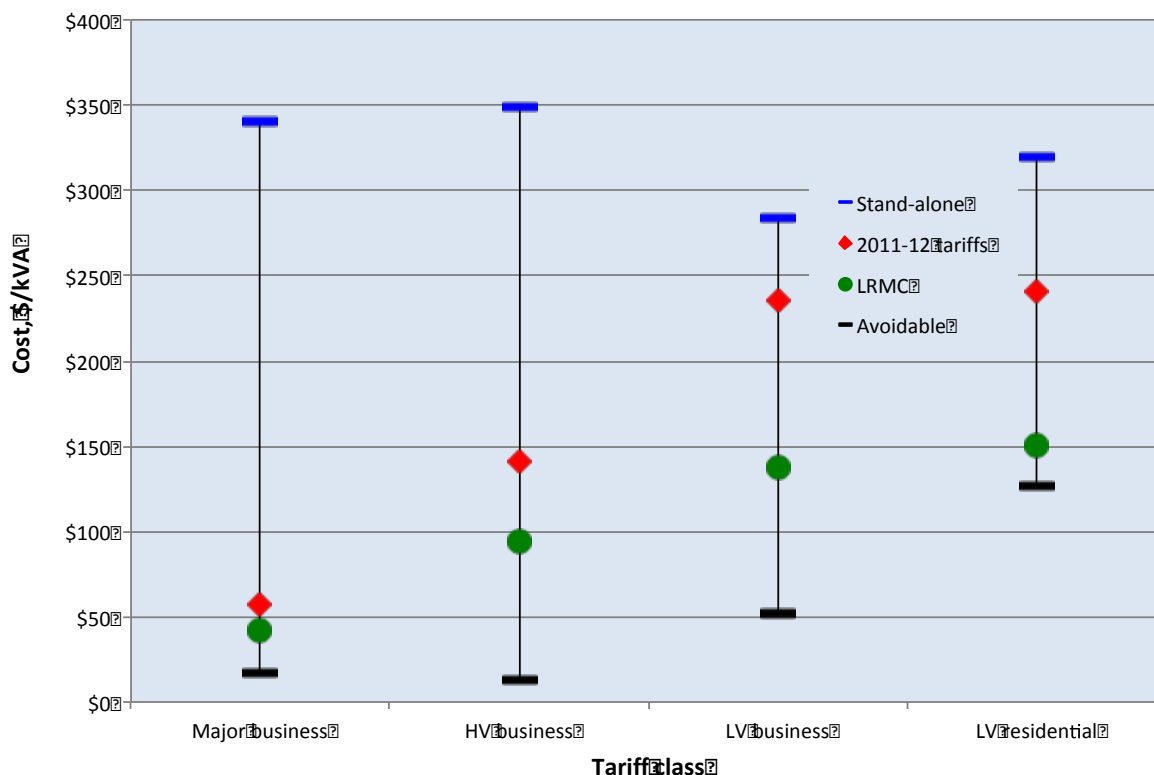
<sup>21</sup> Marsden Jacob, LRM, November 2004, p.16.

<sup>22</sup> EnergyAustralia, Network Pricing Proposal, May 2009.

difference between the LRMV of the Low Voltage Business and Residential tariff classes arises because of their different power factors.

A comparison of the stand-alone, avoidable, LRMV and 2011/12 tariff rates for ETSA Utilities' four tariff classes are shown in Figure 12<sup>23</sup>.

**Figure 12 - Cost comparison (\$/kVA per annum)**



It can be noted from Figure 12 that:

- The 2011/12 prices for each network tariff class fall well within the bounds of the stand-alone and avoidable costs and hence are subsidy-free;
- The LRMV of each tariff class determined from the approach described above yields a cost that is less than that expected to be recovered through the 2011/12 prices in the case of all tariff classes; and
- For the formulation of these estimates, kVA peak demands coincidental with ETSA Utilities' peak were estimated for each tariff. In the case of the low voltage business tariff class, its relatively lower contribution to ETSA Utilities' coincidental peak demand (compared with residential tariff which tends to drive a greater proportion of the overall peak) has resulted in a tariff rate which is higher in \$/kVA terms.

<sup>23</sup> The stand-alone, 2010/11 tariff and avoidable tariff class rates are expressed as their \$ contribution divided by the forecast coincident peak kVA for ETSA Utilities' system demand in 2010/11, with a 10% PoE.

## 8.9.1 Application of the LRM to price formulation

As required by clauses 6.18.5(b)(1) and 6.18.5(c) of the Rules, ETSA Utilities has taken into account the calculated values of LRM, in establishing the charging parameters for each of ETSA Utilities' 2011/12 network tariffs.

Charging parameters of tariffs that are related to volume may be expected to influence customers' consumption decisions. Those parameters are:

- Monthly demand;
- Peak period energy; and
- To a much less significant extent, anytime energy.

On the other hand, charging parameters that are effectively fixed would be unlikely to affect customer consumption patterns, throughout a very broad range.

ETSA Utilities' tariffs for larger business customers have a ToU structure, having demand and energy charges that apply or are higher during a defined peak consumption period. The revenue that is forecast to be recovered in 2011/12 from these peak period business kVA demand and 2-rate tariff components is compared with the LRM for tariff classes in Table 39A and B. This Table also illustrates the proportions of tariff revenue recovered from peak period charges, off-peak charges and fixed charges.

In order to convert the price signalling components of tariffs to the equivalent annual \$/kVA charges in this price comparison, the revenue per annum derived from the application of each price component during peak periods was divided by the kVA demand for the tariff. The associated calculations are described in Appendix E.

**Table 39A - LRM and volume related charging parameters for Time of Use**

|                            | Major Business tariff class |                      |          | HV Business tariff class | LV Business tariff class |         |
|----------------------------|-----------------------------|----------------------|----------|--------------------------|--------------------------|---------|
|                            | Subtrans                    | Zone Sub (local UoS) | Zone Sub |                          |                          |         |
|                            |                             |                      |          |                          |                          |         |
|                            | kVA demand                  |                      |          |                          | 2-rate                   |         |
| LRMC\$/kVA/yr              | \$42                        |                      |          | \$94                     | \$138                    |         |
| Price Signalling           |                             |                      |          |                          |                          |         |
| Peak demand rate\$/kVA/yr  | \$7.60                      | \$17.30              | \$17.30  | \$24.90                  | \$83.20                  |         |
| Peak energy rate\$/kVA/yr  | \$0.50                      | \$1.50               | \$3.10   | \$6.70                   | \$6.70                   | \$34.60 |
| Total peak charge\$/kVA/yr | \$8.10                      | \$18.80              | \$20.40  | \$31.60                  | \$89.90                  | \$34.60 |

- (a) Energy rate is charged in \$/MWh but is expressed here in equivalent \$/kVA terms over the period of peak demand. The revenue recovery forecast by each charge during the peak period is divided by the associated peak kVA demand.

The equivalent LRM and revenue information for ETSA Utilities' single rate energy tariffs is illustrated in Table 39B. For the purpose of estimating an equivalent \$/kVA rate, it was assumed that loads in excess of 80% of the peak demand (which equates to only 1.4% of all hours) contributed to the peak load that drives augmentation of the network.

**Table 39B - LRM and volume related charging parameters for single rate**

|                         | LV Business Tariff Class |           | Residential Tariff Class |
|-------------------------|--------------------------|-----------|--------------------------|
|                         | Single Rate              | Unmetered | Residential-CL           |
| LRM \$/kVA/yr           | \$138                    |           | \$150                    |
| <b>Price Signalling</b> |                          |           |                          |
| Energy Rate \$/kVA/yr   | \$14.00                  | \$25.00   | \$16.00                  |

- (a) Energy rate is charged in \$/MWh but is expressed here in equivalent \$/kVA terms over the period of peak demand. The revenue recovery forecast by each charge during the peak period is divided by the associated peak kVA demand.

The way in which the tariff class LRM has been taken into account by ETSA Utilities in establishing the 2011/12 tariff components has involved the following considerations:

- **Ensuring that price signalling components do not oversignal the LRM:** As customers' consumption decisions will be influenced by the charging parameters of tariffs related to volume, ETSA Utilities has confirmed that, expressed on a common basis, the price signalling tariff components have all been set less than the LRM of supply. The network tariffs will therefore not over-signal the marginal cost of supply;
- **Use of price signalling components where practicable:** Where permitted by the metering arrangements, volume related charges that reflect the customers' demand or incremental energy consumption have been used to improve signalling of the tariff class LRM;
- **Use of stepped pricing signals:** ETSA Utilities' stepped energy rates are being used to signal the costs of network expansion in an incremental manner. With single rate customers the energy rate increases as consumption increases, reflecting the correspondence between customer energy consumption and the proportion of premises that are air conditioned, have poor load factors and impose additional network expansion costs;
- **Revenue recovery through non-distortionary charging parameters:** For each tariff, price signalling charging parameters recover a proportion of the total revenue during peak consumption periods. The balance of revenue recovery takes place in the least distortionary manner possible, through fixed monthly charges and through the single rate energy rates that apply during off peak periods.

ETSA Utilities' 2011/12 tariffs have therefore been established in compliance with the provisions of clauses 6.18.5(b)(1) and 6.18.5(c) of the Rules.

There are other ways in which the LRMC of network expansion is being used by ETSA Utilities to guide its tariff strategy for future years, namely:

- Business customers are being encouraged to take up the kVA demand tariff, which may be seen from Table 39A to more effectively pass on price signals for the use of the network than the single rate or 2-rate or single rate energy tariffs; and
- The indicative future tariff charging parameter price movements set out in Table 27 have been established with a view to enhancing the alignment between ETSA Utilities' network price signals and the cost of supply.

## **8.10 Transaction costs**

Clause 6.18.5(b)(2)(i) of the Rules requires ETSA Utilities to have regard to the transaction costs arising from its network tariffs, by limiting the complexity of tariff structures and the number of charging parameters within each tariff. The charging parameters applicable to each tariff are provided in Chapter 4 of this Pricing Proposal.

ETSA Utilities has not introduced new tariffs or tariff structures in 2011/12 and thus transaction costs will not be increased as a result of the implementation of these prices. ETSA Utilities is therefore compliant with this Rules provision.

## **8.11 Customer response to price signals**

ETSA Utilities is required to have regard to the ability of customers to respond to the price signals provided by its network tariffs, in accordance with clause 6.18.5(b)(2)(ii) of the Rules. The efficiency gains of marginal cost pricing are realised when a tariff based on the marginal cost of supply induces the customer to make behavioural change.

To the extent possible within the limitations imposed by network tariff structures and metering constraints, ETSA Utilities signals the long run marginal cost of supply through those tariff charging parameters with the greatest price elasticity of demand, namely the variable consumption charges that are based on the customers energy use and maximum demand.

In relation to the operation of clause 6.18.5(c) of the Rules, it is noted that ETSA Utilities' current estimate of LRMC falls above the all of the price signalling charging parameters in each tariff class, as described in section 8.9.

If the price signalling charging parameters alone (which were set taking into account the LRMC) were used, the revenue for each tariff class would be insufficient to recover the expected revenue. The revenue shortfall is recovered through the use of tariff components which would cause minimal distortion in efficient patterns of consumption, namely:

- Fixed charges; and
- Anytime energy charges during off peak periods.

ETSA Utilities is therefore compliant with this Rules provision.

## 9 Transmission cost recovery tariffs

This section sets out the procedures that ETSA Utilities will follow to enable the recovery of transmission related charges, including avoided TUoS charges paid to the embedded generators connected to ETSA Utilities' distribution network.

ElectraNet is the Transmission Network Service Provider in South Australia and calculates location specific transmission prices for the connection points to ETSA Utilities' network, in accordance with the provisions of the Rules.

ElectraNet's 2011/12 revenue will again be offset due to revenue from settlement residue auctions. ElectraNet has advised that the amount of settlement residue auction proceeds revenue has fallen significantly, resulting in a loss of almost all of the 'discount' that has previously been applied to SA transmission tariffs (the ElectraNet price discount in 2010/11 was over 10% of full prices).

This section 9 has been prepared using advice from ElectraNet that ETSA Utilities' transmission charges will be increasing by 17.6% this year. This section will not need to be updated if ElectraNet's proposed prices for 2011/12 are approved by the AER.

### 9.1 Regulatory Requirements

#### 9.1.1 Rules requirements

ETSA Utilities' Pricing Proposal is required under clause 6.18.2(b) of the Rules to set out how the TUoS charges it incurs are passed on to customers.

#### 6.18.2 Pricing proposals

(b) A *pricing proposal* must:

- (3) set out, for each proposed tariff, the *charging parameters* and the elements of service to which each *charging parameter* relates; and
- ...
- (6) set out how charges incurred by the *Distribution Network Service Provider* for *transmission use of system services* are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous *regulatory year*;

Clause 6.18.7 of the Rules sets out the requirement concerning the recovery of transmission related charges by ETSA Utilities.

#### 6.18.7 Recovery of charges for transmission use of system services

- (a) A *pricing proposal* must provide for tariffs designed to pass on to customers the charges to be incurred by the *Distribution Network Service Provider* for *transmission use of system services*.
- (b) The amount to be passed on to customers for a particular *regulatory year* must not exceed the estimated amount of the *transmission use of system* charges for the

relevant *regulatory year* adjusted for over or under recovery in the previous *regulatory year*.

- (c) The extent of the over or under recovery is the difference between:
- (1) the amount actually paid by the *Distribution Network Service Provider* by way of *transmission use of system* charges in the previous *regulatory year*; and
  - (2) the amount passed on to customers by way of *transmission use of system* charges by the *Distribution Network Service Provider* in the previous *regulatory year*.

## 9.1.2 Requirements of the AER's Decision

The AER has required ETSA Utilities as part of its Pricing Proposal for each regulatory year of the 2010-15 regulatory control period, to provide the amounts for the following entries in its TUoS unders and overs account for the most recently completed regulatory year, the current regulatory year and the next regulatory year<sup>24</sup>:

1. Opening balance for each year.
2. Interest accrued on the opening balance for each year, calculated at the rate of the post-tax nominal rate of return as approved by the AER in its distribution determination.
3. The amount of revenue recovered from TUoS charges applied in respect of that year, less the amounts of all transmission related payments made by the DNSP in respect of that year.
4. Six months interest on the net amount in item 3, accrued at the approved post-tax nominal rate of return.
5. Summation of the above amounts to derive the closing balance for each year.

In the determination, the AER established the requirement for ETSA Utilities to maintain a TUoS unders and overs account in a specific format. The AER also requires ETSA Utilities to target a zero expected balance on its TUoS unders and overs account in each regulatory year during the 2010-15 regulatory control period.

## 9.2 Transmission cost recovery tariff methodology

The key principles of ETSA Utilities' transmission cost recovery tariff (TCR) methodology are:

- The total TUoS allocated to network tariffs aligns with the total estimated transmission charge to be paid by ETSA Utilities, adjusted for any overs and unders account balance;
- To the extent possible given the constraints of metering and tariff structures, transmission charges are allocated to network tariffs in a manner that reflects the cost drivers present in transmission pricing;
- Customers with a demand of 10 MW or consumption in excess of 40 GWh have individually calculated tariffs with transmission charges allocated in a manner that

<sup>24</sup> AER, Final Decision - South Australia distribution determination 2010-11 to 2014-15, May 2010, pp. 322-323.

preserves the location and time signals of transmission pricing in accordance with the principles in Part J of chapter 6A of the Rules; and

- Network tariffs for smaller customer classes have transmission charges allocated on an energy basis, as location signals cannot be preserved.

## 9.2.1 Avoided TUoS payments

With respect to avoided TUoS for embedded generators, ETSA Utilities calculates the avoided TUoS for all embedded generators that export to its distribution network at the same rates for the locational component which would be applied to a load of similar size at the same connection point.

This payment of avoided TUoS charges to embedded generators is as required under clauses 5.5(h), 5.5(i) and 5.5(j) of the Rules. These avoided TUoS payments to embedded generators are recouped through the recovery mechanism for the TUoS charges.

## 9.3 Transmission use of system overs and unders account balance

In accordance with Appendix F of the 2010 distribution determination, and consistent with the example TUoS calculation provided in Table F.1, Table 40 provides the forecast 2011/12 balance of ETSA Utilities' transmission use of system overs and unders amounts.

**Table 40 - Transmission overs and unders account balance (\$'000)**

|                                                 |        | 2009/10<br>Actual | 2010/11<br>Expected | 2011/12<br>Forecast |
|-------------------------------------------------|--------|-------------------|---------------------|---------------------|
| Revenue from Transmission Cost Recovery charges | \$'000 | 1198,678          | 1220,814            | 1266,581            |
| Transmission charges to be paid to DNSPs        | \$'000 | 1191,397          | 1227,281            | 1267,271            |
| (Settlement residue payments)                   | \$'000 | 0                 | 0                   | 0                   |
| Avoided TUoS payments                           | \$'000 | 0                 | 0                   | 0                   |
| Inter-DNSP payments                             | \$'000 | 0                 | 0                   | 0                   |
| Total transmission related payments             | \$'000 | 1191,397          | 1227,281            | 1267,271            |
| <b>Unders and overs account</b>                 |        |                   |                     |                     |
| Annual rate of interest applicable to balances  | %      | 0.00%             | 9.76%               | 9.76%               |
| Half-year rate of interest                      | %      | 0.00%             | 4.74%               | 4.74%               |
| Opening balance                                 | \$'000 | (509)             | 6,773               | 659                 |
| Interest on opening balance                     | \$'000 | 0                 | 61                  | 64                  |
| Under/over recovery for financial year          | \$'000 | 7,281             | (5,467)             | (590)               |
| Interest on under/over recovery                 | \$'000 | 0                 | (308)               | (33)                |
| Closing balance                                 | \$'000 | 6,773             | 659                 | (0)                 |

The reconciliation in Table 40 demonstrates an immaterial closing balance for the Unders and overs account in 2011/12 and thence ETSA Utilities' compliance with the provisions of clause 6.18.7 of the Rules and the AER's 2010 determination.

## 9.4 Charging parameters for transmission recovery tariffs

ETSA Utilities' transmission recovery tariffs are included in the bundled NUoS Rates of customer tariffs. The charging parameters associated with transmission cost recovery tariffs are shown in sections 4.3 to 4.6, in Table 4 through to Table 14. For customers with a

demand greater than 10 MW or consumption in excess of 40 MWh the transmission cost recovery tariff is location specific; for smaller customers it is averaged.

Transmission cost recovery amounts are billed at the same frequency as the relevant tariff for standard control services.

## **9.5 Transmission recovery tariffs for 2011/12**

ETSA Utilities' 2011/12 recovery through transmission tariffs is forecast to increase from \$221 M in 2010/11, to \$267 M. This is a 20.7% increase, which arises for the following three reasons:

- The approved annual increase in annual ElectraNet revenue under their 2008-2013 Price Reset, and also the recovery of approved Murraylink revenue;
- The 'discount' of over 10% in ElectraNet's charges in 2010/11, due to settlement residues, has been reduced to negligible amounts in 2011/12; and
- The under recovery in pricing targetted by ETSA Utilities in 2011/12 (\$0.7M) is significantly less than in 2010/11 (\$6.5M).

ETSA Utilities has prepared prices for 2011/12 that recover the forecast ElectraNet charges. Locational customers have prices developed directly from the ElectraNet price list according to their location and usage parameters. All other customers have had prices applied on a State-wide non-locational basis, using the pricing signals provided by ElectraNet, the billing parameters available for that customer segment and the customer demand assumptions for that customer segment. For example, business customers on a demand tariff will incur a mixture of demand and energy charges for transmission, whilst residential and small business customers will incur an energy-based charge. The 2010-11 non-locational customer transmission prices have been uniformly escalated by 19.5% for 2011-12.

## 10 Customer tariff class assignment and reassignment

The requirements concerning the assignment and reassignment of customer to tariff classes are set out in clause 6.18.4 of the Rules and Chapter 2 and Appendix B of the AER's Decision.

### 10.1 Regulatory Requirements

#### 10.1.1 Rules requirements

In making a distribution determination, the AER is required to formulate provisions for the assignment and reassignment of customers to tariff classes, in accordance with the principles set out in clause 6.18.4 of the Rules. This Rule covers the following matters:

- Factors governing the assignment of customers to tariff classes;
- Equitable treatment of customers with micro-generation;
- The review of the DNSP decision on tariff class assignment; and
- The review of DNSPs' tariff structures containing energy or demand related charges.

#### 10.1.2 Requirements of the AER's Decision

In accordance with the principles in clause 6.18.4 of the Rules, Appendix B of the AER's decision sets out the procedures to apply to assigning or reassigning customers to tariff classes<sup>25</sup>. These provisions are in several parts, covering the following aspects:

- The initial assignment of customers at the commencement of the 2010-15 regulatory control period;
- Assignment of new customers to a tariff class during the next regulatory control period;
- Reassignment of existing customers to another existing or a new tariff during the next regulatory control period;
- Objections to proposed assignments and reassignments; and
- System of assessment and review of the basis on which a customer is charged.

The initial assignment of existing standard control services customers to their existing tariffs was discussed in section 4.1.2. The remaining elements of the AER's Decision on tariff assignment and reassignment are set out below.

#### Assignment of new customers to a tariff class during the next regulatory control period

2. If, after 1 July 2010, ETSA Utilities becomes aware that a person will become a customer, then ETSA Utilities must determine the tariff class to which the new customer will be assigned.

<sup>25</sup> AER, Final Decision, May 2010,, pp. 286-288.

3. In determining the tariff class to which a customer or potential customer will be assigned, or reassigned, in accordance with section 2 or 5, ETSA Utilities must take into account one or more of the following factors:
  - (a) the nature and extent of the customer's usage
  - (b) the nature of the customer's connection to the network<sup>26</sup>
  - (c) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement.
4. In addition to the requirements under section 3, ETSA Utilities, when assigning or reassigning a customer to a tariff class, must ensure the following:
  - (a) that customers with similar connection and usage profiles are treated equally
  - (b) that customers which have micro-generation facilities are not treated less favourably than customers with similar load profiles without such facilities.

### **Reassignment of existing customers to another existing or a new tariff during the next regulatory control period**

5. If ETSA Utilities believes that an existing customer's load characteristics or connection characteristics (or both) have changed such that it is no longer appropriate for that customer to be assigned to the tariff class to which the customer is currently assigned or a customer no longer has the same or materially similar load or connection characteristics as other customers on the customer's existing tariff, then it may reassign that customer to another tariff class.

### **Objections to proposed assignments and reassignments**

6. ETSA Utilities must notify the customer concerned in writing of the tariff class to which the customer has been assigned or reassigned by it, prior to the assignment or reassignment occurring. If ETSA Utilities does not know the identity of the customer then it must notify the customer's retailer instead.
7. The notice under section 6 must include advice that the customer may request further information from the DNSP and that it may object to the proposed assignment or reassignment. This notice must specifically include:
  - a. either a copy of ETSA Utilities internal procedures for reviewing objections or the link to where such information is available on ETSA Utilities' website
  - b. that if the objection is not resolved to the satisfaction of the customer under ETSA Utilities' internal review system, then to the extent that resolution of such disputes are within the jurisdiction of a state based energy Ombudsman scheme the customer is entitled to escalate the matter to such a body
  - c. that if the objection is not resolved to the satisfaction of the customer under the DNSP's internal review system, then the customer is entitled to seek resolution via the dispute resolution process available under Part 10 of the NEL.

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<sup>26</sup> The AER interprets 'connection' to include the installation of any technology capable of supporting time based tariffs.

8. If, in response to a notice issued in accordance with section 6, ETSA Utilities receives a request for further information from a customer, then it must provide such information. If any of the information requested by the customer is confidential then it is not required to provide that information to the customer.
9. If, in response to a notice issued in accordance with section 7, a customer makes an objection to ETSA Utilities about the proposed assignment or reassignment, ETSA Utilities must reconsider the proposed assignment or reassignment, taking into consideration the factors in sections 3 and 4 above, and notify the customer in writing of its decision and the reasons for that decision.
10. If a customer's objection to a tariff assignment or reassignment is upheld by the relevant external dispute resolution body, then any adjustment which needs to be made to prices will be done by ETSA Utilities as part of the next annual review of prices.

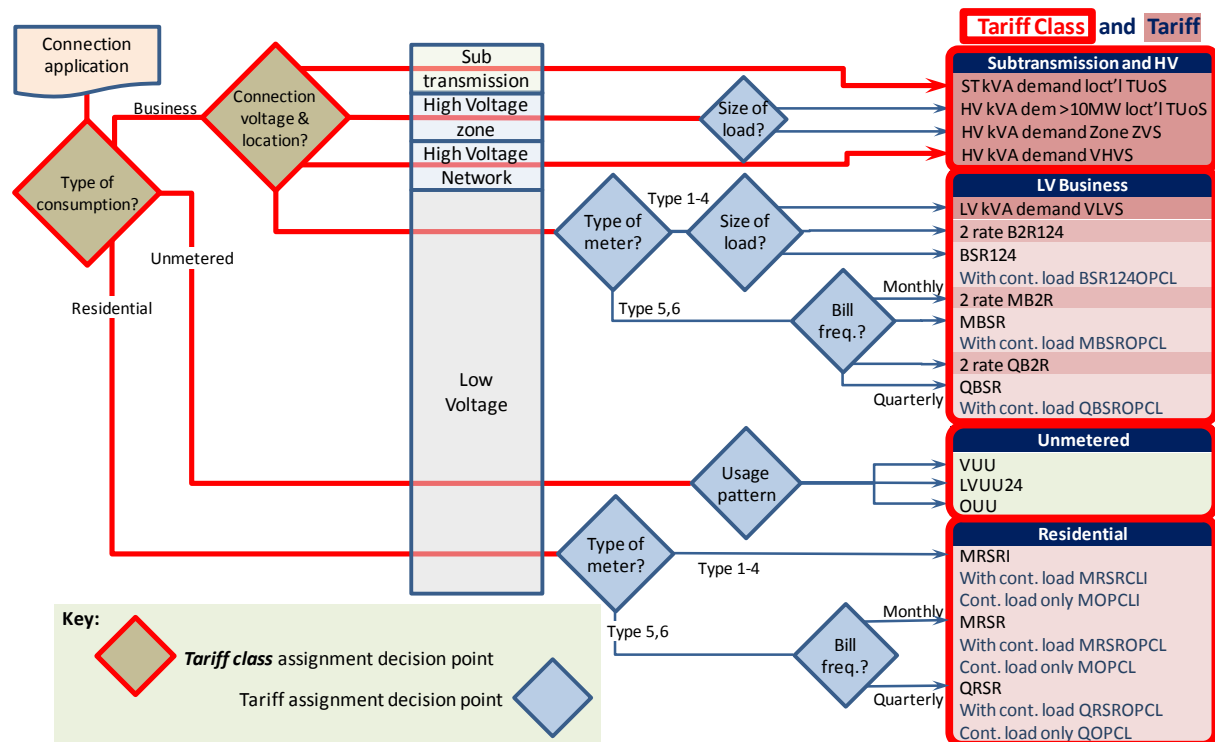
## **10.2 Assignment of new customers to a tariff class during the next regulatory control period**

The approach that ETSA Utilities applies to the tariff assignment of new and upgraded customer connections has been developed over the years since the formation of the NEM.

In this section of the Pricing Proposal, ETSA Utilities describes the process it applies to the initial assignment of customers to tariffs and to their reassignment. Notwithstanding that the individual tariffs have been grouped within tariff classes in this Pricing Proposal, the existing approach to managing tariff assignment and reassignment is demonstrated to align with the requirements established by the AER. Accordingly, no change is required to current practices.

The process whereby new customers are assigned to tariff classes and tariffs, following the receipt of a connection application by the customer or their retailer, follows the decision tree shown in Figure 13. In the application of this process, a customer that lodges an application to modify or upgrade an existing network connection is treated in the same manner as a new customer.

**Figure 13 – Assignment of new and upgraded customer connections to tariff classes**



This decision tree in Figure 13 highlights the existing process whereby customers are assigned to a tariff class and then to an individual tariff. The process relies upon a systematic sequence of decisions based on the information provided with the customer's application for supply. Decisions associated with assignment to the four tariff classes have been separately identified in red. Second-order decisions on individual tariffs are also shown. These relate to type of meter, load size and billing frequency, and lead to the customer's assignment to a specific tariff within the tariff class.

The two major decisions that determine the tariff class assessment are as follows:

- The nature of a customer's usage: (ie. residential, business, or unmetered); and
- For business customers only, the nature and extent of the associated connection to the network (the connection voltage, whether located within in the network or directly connected to a zone substation).

The process employed by ETSA Utilities therefore appropriately takes account the factors in clause 3(a) and 3(b) of the AER's Appendix B.

Moreover, in the event that remotely-read interval metering or other similar metering technology is installed at the customer's premises as a result of a regulatory obligation or

requirement during the 2010-15 regulatory control period, ETSA Utilities will review this procedure.

### **10.2.1 Customers with micro-generation**

As ETSA Utilities' tariff class assignment process is applied to the *net* customer demand on the network, it does not distinguish between customers that have micro-generation and those without. The only aspects of the connection process that distinguish customers with micro-generation are technical requirements, principally to ensure public and employee safety in the event of disconnection of supply to a site with generation.

ETSA Utilities' tariff assignment process therefore ensures that the requirements in clause 4(a) and 4(b) of the AER's Appendix B are met.

### **10.3 Reassignment of existing customers to another existing or a new tariff during the next regulatory control period**

Within each tariff class, there has been and will continue to be movement between individual tariffs. This is particularly the case with the customers on the Low Voltage Business tariff class. Whilst there has been no active review process by ETSA Utilities to ensure that customers whose consumption and usage profiles change are maintained on the most advantageous tariff, customers are eligible to apply for transfer between tariffs and do so if it is to their advantage. This has been the case with business customers that have transferred from the inclining block tariff to capacity based tariffs and between different capacity-based tariffs. ETSA Utilities considers that preserving this level of flexibility to permit customers the option of transferring to a tariff more appropriate to their operations within a tariff class is of great importance to customers.

The four tariff classes that ETSA Utilities has established are sufficiently broad to ensure that all the existing customers are within the appropriate tariff class and that it is unlikely that customers will seek to migrate or be reclassified to a different tariff class during the course of the determination. Transfer between tariff classes would be limited to circumstances where the nature of usage or level of consumption changed significantly, for example where a residence was redeveloped to become a small business such as a medical surgery or office.

Notwithstanding that the reassignment of customers' tariff classes is unlikely during the 2010-15 regulatory control period, ETSA Utilities would do so in accordance with the provisions of the AER's Decision, in particular clause 5 of the Appendix B.

#### **10.3.1 Obsolete tariffs**

In addition to the current tariffs illustrated in Figure 13, in common with most utilities, at the time of the 2010 determination ETSA Utilities had a range of obsolete business tariffs that were all within the two business tariff classes. No new or modified customer connections were subsequently assigned to these obsolete tariffs. As the opportunity arose, customers were transferred from obsolete tariffs to current tariffs within the same tariff class, and by 2010/11, all customers were transferred from the obsolete kW demand tariffs.

When a customer was transferred from an obsolete tariff to one of the current tariffs, the choice of the appropriate tariff followed the tariff assignment decision process in section 10.2 of this Pricing Proposal.

The formulation of the WAPC requires that the reasonable estimates provisions be used to create equivalent historical quantities for these customers in the 2009/10 year. These reasonable estimates for kW demand are detailed in Appendix I (Confidential).

#### **10.4 Objections to proposed assignments and reassignments**

The AER has established requirements that ETSA Utilities must follow in assigning or reassigning customers to tariff classes and in responding to objections to ETSA Utilities' tariff class assignments. These are set out in the Decision as clauses 7 to 10 of the AER's Appendix B.

The requirements that ETSA Utilities must follow have been documented in an internal procedure entitled "Manual 18, Network Tariff Manual".

This document is attached as to this Pricing Proposal as Appendix K and is published on ETSA Utilities' web site.

## 11 Alternative Control Services – metering services

In the 2010 Decision, in Appendix A the AER has classified the following distribution services provided by ETSA Utilities as alternative control services<sup>27</sup>:

### A.5 ‘Variable’ ‘standard’ ‘small’ customer metering services

- a. The provision of:
  - i. meter provision services in respect of meters meeting the requirements of a *metering installation* type 6, and
  - ii. quarterly meter read services in respect of meters meeting the requirements of a *metering installation* type 6.
- b. For the purposes of this clause, meter provision services include, but are not necessarily limited to, any asset related and administrative costs associated with the provision, installation, maintenance, and replacement of the meter (including circumstances in which ETSA Utilities meter is replaced by that of another meter provider).

### A.6 ‘Exceptional’ large customer metering services

- a. Meter provision services provided in respect of meters meeting the requirements of a *metering installation* type 1, *metering installation* type 2, *metering installation* type 3 or *metering installation* type 4 installed prior to 1 July 2000.
- b. Meter provision services provided in accordance with the requirement of clause 27 of ETSA Utilities’ distribution licence as in force at 30 June 2005.
- c. For the purposes of this clause, meter provision services include, but are not necessarily limited to, any asset related and administrative costs associated with the provision, installation, maintenance, and replacement of the meter (including circumstances in which ETSA Utilities meter is replaced by that of another meter provider).

This section of the Pricing Proposal sets out ETSA Utilities’ approach to the pricing of its alternative control services and demonstrates compliance with the Rules and the AER’s determination.

### 11.1 Regulatory requirements

#### 11.1.1 Rule requirements

The Rule requirements pertaining to Pricing Proposals that apply to direct control services are applicable to both standard control services and alternative control services.

<sup>27</sup> AER, Final Decision – South Australia distribution determination 2010–11 to 2014–15, May 2010, p. 281.

The Rule requirements applicable to standard control services have been set out in various earlier sections of this Pricing Proposal. For brevity, these clauses have not been repeated in this section on alternative control services. The relevant Rule clauses are as follows:

**6.18.2 Pricing proposals** - section 4.1.1, page 14.

**6.18.3 Tariff classes** - section 4.1.1, page 14.

**6.18.4 Principles governing assignment or re-assignment of customers to tariff classes and assessment and review of basis of charging** - section 10.1, page 70.

**6.18.5 Pricing principles** – section 8.5, page 58.

## **11.1.2 Requirements of the AER's Decisions**

The AER's Draft and Final Decisions imposed the following requirements on ETSA Utilities in respect of alternative control services<sup>28, 29</sup>:

1. In accordance with clause 6.12.1(12) of the NER, the control mechanism for the alternative control services provided by ETSA Utilities is a weighted average price cap. The applicable WAPC formula was set out in section 17.3 of the draft AER's draft decision. ETSA Utilities is required to include proposed distribution tariff classes (n) and components (m) for both variable standard small customer metering services and the two exceptional case metering services.
2. In accordance with clause 6.12.1(13) of the NER, ETSA Utilities must demonstrate compliance with the control mechanism for alternative control services by providing, as part of its annual pricing proposal, the proposed tariffs which correspond to the price terms contained in the WAPC equation.

Consistent with requirement 1, ETSA Utilities' Revised Proposal incorporated one alternative control services tariff class for metering services comprising six meter provision tariffs. The AER accepted that the approach ETSA Utilities used to develop the meter tariffs of the metering services tariff class was reasonable.<sup>30</sup>

Consistent with requirement 2, ETSA Utilities' Revised Proposal stated that ETSA Utilities would demonstrate compliance with the control mechanism by annually providing the proposed tariffs that correspond to the price terms contained in the WAPC approved by the AER. The AER accepted this proposal.<sup>31</sup>

The AER accepted ETSA Utilities' proposal that the alternative control services X factor for each year of the next regulatory control period be made equal to deliver a smooth price path within the next regulatory control period.

<sup>28</sup> AER, draft Decision – South Australia distribution determination 2010–11 to 2014–15, 25 November 2009, pp. 422, 427.

<sup>29</sup> AER, Final Decision – South Australia distribution determination 2010–11 to 2014–15, May 2010, p. 274.

<sup>30</sup> AER, Final Decision, May 2010, pp. 258-261, 266.

<sup>31</sup> AER, Final Decision, May 2010, p. 273.

The AER also accepted ETSA Utilities' proposal to initially assign all metering service tariffs to a single alternative control tariff class<sup>32</sup>.

The AER's Final Decision confirmed that the Weighted Average Price Cap form of price control detailed in section 2.3.4 of this Pricing Proposal would apply to ETSA Utilities' alternative control metering services. The AER determined a constant X factor of -8.05% for the 2010-15 regulatory control period, as detailed in section 2.3.5.

Appendix B of the AER's Final Decision sets out the procedures for assigning and reassigning customers to tariff classes. ETSA Utilities assumes these procedures apply, where relevant, to both standard control and alternative control services.

## **11.2 Alternative control metering services tariff class**

Clause 6.18.3(c) of the Rules requires ETSA Utilities to constitute separate tariff classes for customers to whom standard control services are supplied and customers to whom alternative control services are supplied.

ETSA Utilities' tariff classes for standard control services are described in section 10.1.2 of this Pricing Proposal.

Clause 6.18.3(d) of the Rules requires ETSA Utilities, in constituting tariff classes, to have regard for the need to group customers together on an economically efficient basis and avoid unnecessary transaction costs.

Consistent with these requirements, ETSA Utilities has constituted a single separate tariff class for its alternative control metering services. This tariff class comprises all six alternative control metering service tariffs. These are set out in Table 41.

## **11.3 Assignment and reassignment of customers to the alternative control metering service tariff class**

ETSA Utilities has assigned all of its metering service tariff customers to a single alternative control metering service tariff class. Any new metering services customer during the 2010-15 regulatory control period will be assigned to this tariff class.

As there is only a single tariff class proposed, there will be no requirement to reassign customers to another alternative control tariff class during the 2010-15 regulatory control period.

## **11.4 Pricing principles**

Clause 6.18.5 of the Rules sets out the pricing principles that must be complied with in respect of each tariff class, including a tariff class within the classification of alternative control services.

As noted in section 11.2, ETSA Utilities has established a single tariff class for its alternative control metering services.

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<sup>32</sup> AER, Final Decision, May 2010, p. 257.

## **11.4.1 Stand alone and avoidable costs of alternative control services**

Rule 6.18.5(a) requires the revenue of each tariff class to lie on or between the stand-alone and avoidable costs of serving the customers in the tariff class.

As described in ETSA Utilities' Revised Proposal, the methodology used to determine the cost relativities in respect of each of the tariffs<sup>33</sup> within the metering services tariff class involved the determination of the variable cost of providing the metering services relevant to each tariff.<sup>34</sup> These variable cost reflective tariff charges were applied to estimated customer numbers of the relevant tariffs to ensure the recovery of expected revenue, as set out in ETSA Utilities' Revised Proposal. This methodology delivers revenue from the alternative control services metering services tariff class that reflects the cost that would be avoided by not serving those customers.

Furthermore, given that 99.9% of alternative control services customers are subject to variable standard small customer metering services, stand-alone costs have been assessed as being approximately equal to the revenue from the alternative control services metering services tariff class.

ETSA Utilities' metering tariff class therefore meets the requirements of Rule 6.18.5(a).

## **11.4.2 Long Run Marginal Costs and revenue recovery**

Rule 6.18.5(b) requires each charging parameter for a tariff class to take into account the LRMC of providing that service.

The long-run marginal cost is the change in the long-run total cost of providing the service resulting from a change in the quantity of services provided. There are no fixed inputs in the long run, only variable costs. This means that long-run marginal cost is the result of changes in the cost of all inputs, or (in long-run equilibrium at least) the equivalent of the long-run average total cost. From a practical perspective, long-run marginal cost can be disaggregated into two types of marginal costs, marginal operating costs (generally with short-run characteristics) and marginal capacity costs (exhibiting long-run characteristics).<sup>35</sup>

As required by the AER's classification of alternative control services metering services, ETSA Utilities' determination of total alternative control services metering services costs for the next regulatory control period was undertaken on a solely variable cost basis. Having undertaken a variable cost based building block approach to determining the total variable costs (and thus revenue requirement) for the next regulatory control period, ETSA Utilities' operating expenditure divided by 'output' therefore reflects marginal operating costs, and its asset costs divided by 'output' reflect marginal capacity costs - together the long-run marginal cost. ETSA Utilities has thus taken into account the long-run marginal cost in the development of the metering services tariffs.

<sup>33</sup> Referred to as tariff components in the Revised Proposal.

<sup>34</sup> While only 'Variable' 'standard' 'small' customer metering services are variable by definition, the analysis of the costs associated with Exceptional large customer metering services was undertaken in the same way.

<sup>35</sup> Marsden Jacob Associates "Estimation of Long Run Marginal Cost (LRMC)" 3 November 2004, p.10 (<http://www.qca.org.au/files/QCALRMCFinal.pdf>)

As noted in ETSA Utilities Revised Proposal, the tariffs for metering services were determined having regard to the variable transaction costs associated with the services relevant to each tariff (by virtue of minimising the number of unique tariffs). As noted by the AER in the Final Determination, ETSA Utilities created tariffs to ensure that the tariffs relevant to customers most likely to respond to price signals – the type 6 CT and type 1-4 exceptional meter customers – are explicitly cost reflective.<sup>36</sup>

Thus the requirements of Rule 6.18.5(b)(1) and (2) have been satisfied.

ETSA Utilities' metering tariffs each have a single charging parameter that recovers the whole of the expected revenue. As a consequence, Rule 6.18.5(c) is not applicable.

## 11.5 Compliance with the AER determination

The WAPC equation applicable to ETSA Utilities' alternative control services metering services tariff class for the next regulatory control period is set out in section 17.4 of the AER's Final Decision. This determination was subsequently varied by the Australian Competition Tribunal, as explained in Section 2.3.5. In establishing the proposed 2011/12 prices for alternative control services, ETSA Utilities has used the terms in the WAPC price control formula set out in Table 41.

**Table 41 – Alternative Control Services WAPC price terms**

| Criterion                             | 20010/11 value |
|---------------------------------------|----------------|
| Consumer Price Index                  | 3.33%          |
| X Factor                              | -16.50%        |
| $WAPC = (1 + CPI) \times (1 - X) - 1$ | 20.38%         |

It should be noted that the WAPC form of price control relies upon audited volumes for the year 2009/10 year ( $t-2$ ), in order to establish the prices in 2011/12 year ( $t$ ). In this second year of the 2010-15 regulatory control period, audited volumes will not become available until 2012. ETSA Utilities has substituted the 2008/09 volumes used in the 2010 pricing proposal and also used by the AER, to establish the initial metering tariffs in the WAPC.

ETSA Utilities has demonstrated compliance with the control mechanism for alternative control services by providing, as part of this Pricing Proposal, the proposed tariffs that correspond to the price terms contained in the WAPC equation<sup>37</sup>. These are set out in the WAPC model for alternative control services in Appendix H (Confidential).

<sup>36</sup> AER, Final Decision – South Australia distribution determination 2010–11 to 2014–15, May 2010, p.259

<sup>37</sup> AER, Final Decision, May 2010, p. 274.

Table 42 sets out the tariffs that correspond to the price terms contained in the alternative control services metering services WAPC equation.

**Table 42 – Alternative control services tariffs for 2011/12**

| Tariff   |                                     | Existing<br>Rate<br>2010/11 | Proposed<br>Rate<br>2011/12 | Variance |
|----------|-------------------------------------|-----------------------------|-----------------------------|----------|
| Type 6   | DC Meter Provision \$/day           | 0.0585                      | 0.0706                      | 20.73%   |
| Type 6   | CT Meter Provision \$/day           | 0.2536                      | 0.3062                      | 20.73%   |
| Type 1-4 | Exceptional Meter Provision \$/day  | 0.9005                      | 1.0871                      | 20.73%   |
| Type 1-4 | Other Meter Provider Service \$/day | -                           | -                           | -        |
| Type 6   | Meter Exit Fee Service \$           | 238.70                      | 246.66                      | 3.33%    |
| Type 1-4 | Meter Exit Fee Service \$           | 532.96                      | 550.73                      | 3.33%    |

It should be noted that the change in price for basic the Type 6 meter provision, which applies to the majority of small customers, equates to \$4.43 per annum. As described in section 8.4.2, this has been taken into account in establishing the price for the supply rate of small customer, to ensure that the side constraint on the increase in the fixed charge, of \$10.00, is not breached.

Table 42 also satisfies the requirements of Rules 6.18.2(b)(1), (2), and (3), given that it sets out the tariff classes that are to apply for the 2011/12 regulatory year, the proposed tariffs for each tariff class, and for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

#### **11.6 Charging parameters for alternative control services metering tariffs**

There are only two charging parameters within the alternative control services metering services tariff class: customer numbers; and exit fee transactions.

Meter provision services are charged to each alternative control services network customer on a \$/day basis, so the relevant charging parameter is the number of customer days. Meter services exit fee transactions will be charged on an as incurred basis, so the relevant charging parameter is the number of exit fee transactions. The charging parameters for each tariff within the alternative control services metering services tariff class are set out in Table 43.

**Table 43 – Metering services tariff class charging parameters**

| Tariff class      | Tariff                                               | Charging Parameter      |
|-------------------|------------------------------------------------------|-------------------------|
| Metering services | Meter provision type 6 DCC (\$/day)                  | Number of customer days |
|                   | Meter provision type 6 CTC (\$/day)                  | Number of customer days |
|                   | Meter provision Type 1–4 Exceptional (\$/day)        | Number of customer days |
|                   | Meter service other meter provider customer (\$/day) | Number of customer days |
|                   | Meter service exit fee type 6 CTC (\$)               | Number of transactions  |
|                   | Meter service exit fee type 1–4 (\$)                 | Number of transactions  |

**Appendices**

- Appendix A. 2011-12 Network Use of System Tariffs and explanatory notes**
- Appendix B. CONFIDENTIAL - Audit of EU 2009-10 Quantities by PKF**
- Appendix C. CONFIDENTIAL – WAPC variables 2009-10 including EDPD factor**
- Appendix D. CONFIDENTIAL – ESCOSA verification of EDPD factor**
- Appendix E. Long Run Marginal Cost Methodology**
- Appendix F. Stand-alone and Avoided Cost Methodologies**
- Appendix G. CONFIDENTIAL – AER Weighted Average Price Cap Compliance Model 2011-12 (standard control)**
- Appendix H. CONFIDENTIAL – AER Weighted Average Price Cap Compliance Model 2011-12 (alternate control)**
- Appendix I. CONFIDENTIAL – Reasonable Estimates**
- Appendix J. ETSA Utilities procedure for assigning and reassigning customers to tariff classes (Tariff Manual)**
- Appendix K. Regulatory compliance checklist**